



AutoCAD
ELECTRICAL

COURSE SLIDES · WSQ

Technical Drawing with AutoCAD Electrical

WSQ Course Code: TGS-2023037466

TSC: Technical Drawing · DSN-TDR-4005-1.1

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version 6.0 · 18 August 2026

Trainer: _____ · www.tertiarycourses.com.sg · enquiry@tertiaryinfotech.com · +65 6100 0613

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COURSE ADMINISTRATION

Welcome & Housekeeping

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE

Principal Trainer, Tertiary Infotech Academy Pte Ltd

QUALIFICATIONS

PhD — data science, engineering design and CAD/CAE practice.

DELIVERS

WSQ courses on technical drawing (AutoCAD, AutoCAD Electrical, Revit, Fusion 360) and engineering software.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Let's Know Each Other

- Your name and organisation / role.
- Your experience with AutoCAD or electrical design (if any).
- What you want to be able to draw or document after this course.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is stupid.

3

Respect each other's views.

4

Exit silently for calls or toilet breaks.

5

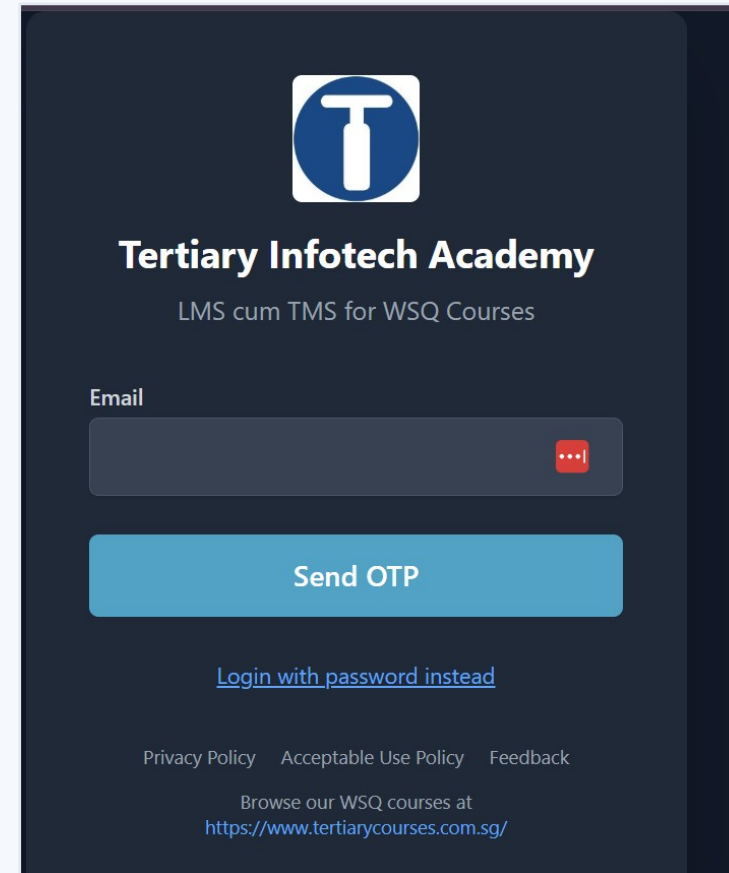
Be punctual; return from breaks on time.

6

75% attendance is required for funding.

Download Course Material

- All course materials live on the Tertiary Infotech LMS/TMS portal.
- Portal: <https://lms-tms.tertiaryinfotech.com>
- Log in with the credentials sent to your registered email.
- Download the slides and Learner Guide — you may use them during the open-book assessment.
- Your attendance records, assessment submission and certificate are also on the portal.



The screenshot shows the login interface for the Tertiary Infotech Academy. At the top is a logo consisting of a white 'T' inside a blue circle. Below the logo, the text 'Tertiary Infotech Academy' is displayed in a bold, white font, followed by 'LMS cum TMS for WSQ Courses' in a smaller, lighter font. The main section features an 'Email' label above a dark grey input field. To the right of the input field is a red button with three white dots. Below the input field is a large, light blue button labeled 'Send OTP'. Underneath this button is a link that says 'Login with password instead'. At the bottom of the page, there are three links: 'Privacy Policy', 'Acceptable Use Policy', and 'Feedback'. Finally, there is a line of text that says 'Browse our WSQ courses at' followed by the URL 'https://www.tertiarycourses.com.sg/'.

Lesson Plan — 2 Days

Day 1

- Day 1 — Getting started, projects and schematic drawing
 - Digital attendance (AM) · introductions · learning outcomes · course outline
 - Topic 1: Introduction to Technical Drawing Using AutoCAD Electrical (Labs 1–2)
 - Lunch break · digital attendance (PM)
 - Topic 2: Analysing Technical Drawings (Labs 3–4)
 - End of Day 1

Day 2

- Day 2 — Analysis, documentation, submission and assessment
 - Digital attendance (AM) · Day 1 recap
 - Topic 2 (continued): cross-referencing & review (Lab 5)
 - Topic 3: Reviewing Drawings for Strategic Relevance & Standard Compliance (Labs 6–8)
 - Course feedback & TRAQOM survey · digital attendance (Assessment)
 - Final Assessment: WA (SAQ) 1 hr + PP 1 hr

Skills Framework

- TSC Title: Technical Drawing
- TSC Code: DSN-TDR-4005-1.1
- This course is mapped to the Skills Framework for Design — Technical Drawing TSC.
- The assessment covers every Knowledge (K1–K6) and Ability (A1–A6) statement of the TSC.

TSC Knowledge Statements

1

K1

Symbols, standards and conventions used in technical drawings across various fields and regions

2

K2

Methods of technical drawing by instruments and computer-aided design (CAD) systems

3

K3

Types of technical drawings

4

K4

Technical drawing standards

5

K5

Fundamentals of drafting design documentation

6

K6

Concepts of information technology (IT) to facilitate the usage of technical drawing software

TSC Ability Statements

1

A1

Devise metrics and guidelines for the production of technical drawings

2

A2

Formulate solutions for problems encountered during the development of technical drawings

3

A3

Review technical drawings for accuracy and adherence to established specifications

4

A4

Analyse drawings to determine their construction needs to deliver upon design solutions

5

A5

Review technical drawings and/or changes to them, to ensure relevance to the organisation's strategies

6

A6

Review drawings and specifications to ensure conformance to requirements of relevant standards for submission

Learning Outcomes

1

LO1 — Devise & solve

Devise metrics for technical drawing production with AutoCAD Electrical and formulate solutions for development problems. (K1, K2, A1, A2)

2

LO2 — Analyse & review

Analyse technical drawings to determine construction needs while reviewing accuracy and adherence to specifications. (K3, K4, A3, A4)

3

LO3 — Strategy & standards

Review drawings for relevance to the organisation's strategies and conformance to submission standards. (K5, K6, A5, A6)

Course Outline

Topic 1 · Introduction

- AutoCAD Electrical user interface
- Symbols, standards & conventions
- Ribbon, palettes & navigation
- Projects & drawing management
- Labs 1–2

Topic 2 · Analysing Drawings

- Types of schematic drawings
- Components, terminals & PLC symbols
- Wires, numbers & signal arrows
- Cross-referencing & review tools
- Labs 3–5

Topic 3 · Review & Compliance

- Drafting design documentation
- Panel layouts & part-list reports
- Custom symbols
- Plot & publish for submission
- Labs 6–8

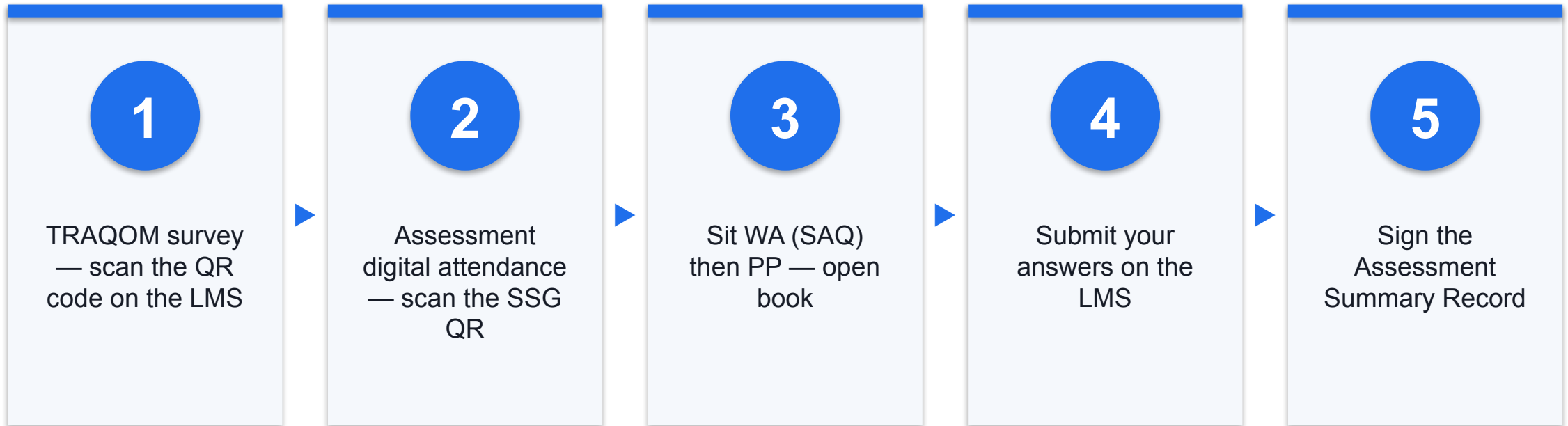
Briefing for Assessment

- Place phones and other materials under the table or on the floor.
- No photos or recording of assessment scripts.
- No discussion during the assessment.
- Use a black/blue pen for hard-copy assessments.
- No liquid paper / correction tape.
- Scripts are collected when time is up.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book. (K1–K6)
- Practical Performance (PP) — hands-on AutoCAD Electrical tasks, 1 hour, open book. (A1–A6)
- Format: Open Book — slides, Learner Guide and approved materials only.
- A minimum of 75% attendance is required to be eligible for assessment and funding.
- An appeal process is available if required.

Assessment Flow



Criteria for Funding

- Minimum attendance rate of 75% based on the SSG Digital Attendance record.
- Complete the assessment and be assessed as 'Competent'.

TOPIC 01

Introduction to Technical Drawing Using AutoCAD Electrical

User interface · symbols, standards & conventions · ribbon · navigation · projects & drawings (K1, K2 · A1, A2)

Key Concepts — Topic 1

1

Symbols, standards, conventions

IEEE 315/315A, IEC-60617 and NFPA symbol libraries make an electrical drawing readable across fields and regions. (K1)

2

CAD vs manual drawing

AutoCAD Electrical automates tagging, wire numbering and cross-referencing that instrument drawing does by hand. (K2)

3

The AutoCAD Electrical UI

Ribbon tabs, Quick Access Toolbar, marking menu, palettes, status bar, command line and navigation tools.

4

Projects drive everything

A .wdp project file groups interrelated drawings so project-wide functions can retag, renumber and report.

5

Devise metrics & guidelines

Templates, drawing properties, reference systems and title blocks standardise production. (A1)

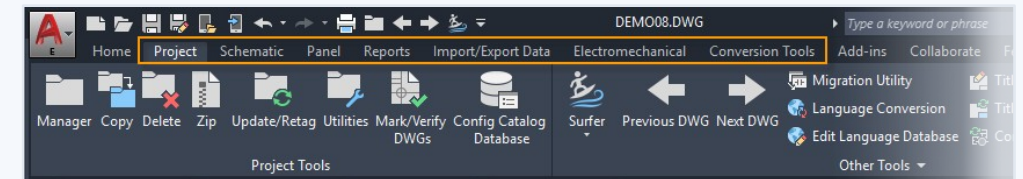
6

Formulate solutions

Grid/snap, ortho, object snap and the Circuit Clipboard solve day-to-day drawing development problems. (A2)

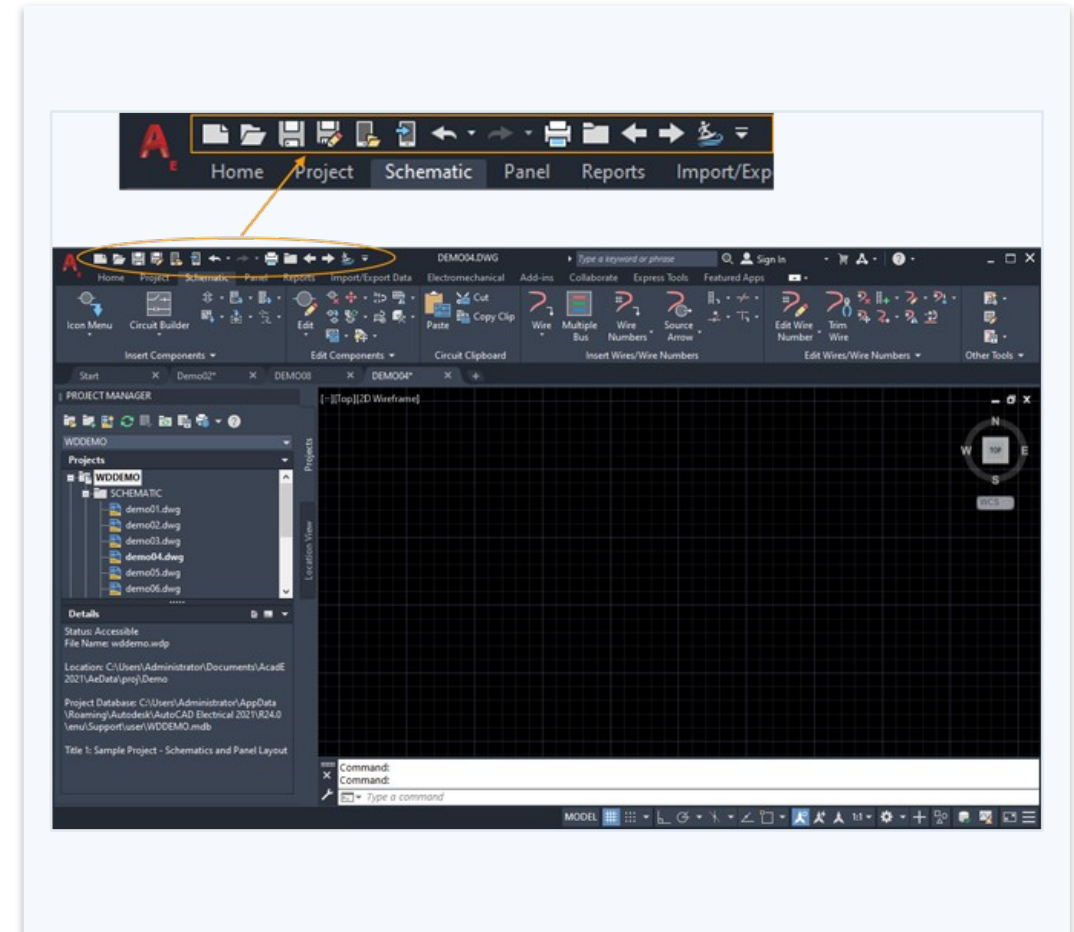
Ribbon Tabs and Panels

- AutoCAD Electrical presents a standard tabbed ribbon across the top of the drawing area.
- Electrical-specific tabs (Schematic, Panel, Reports) sit alongside the core AutoCAD tabs.
- Each tab is divided into panels that group related commands.
- Right-click in the tabs area to choose which tabs and panels to display or hide.



Quick Access Toolbar

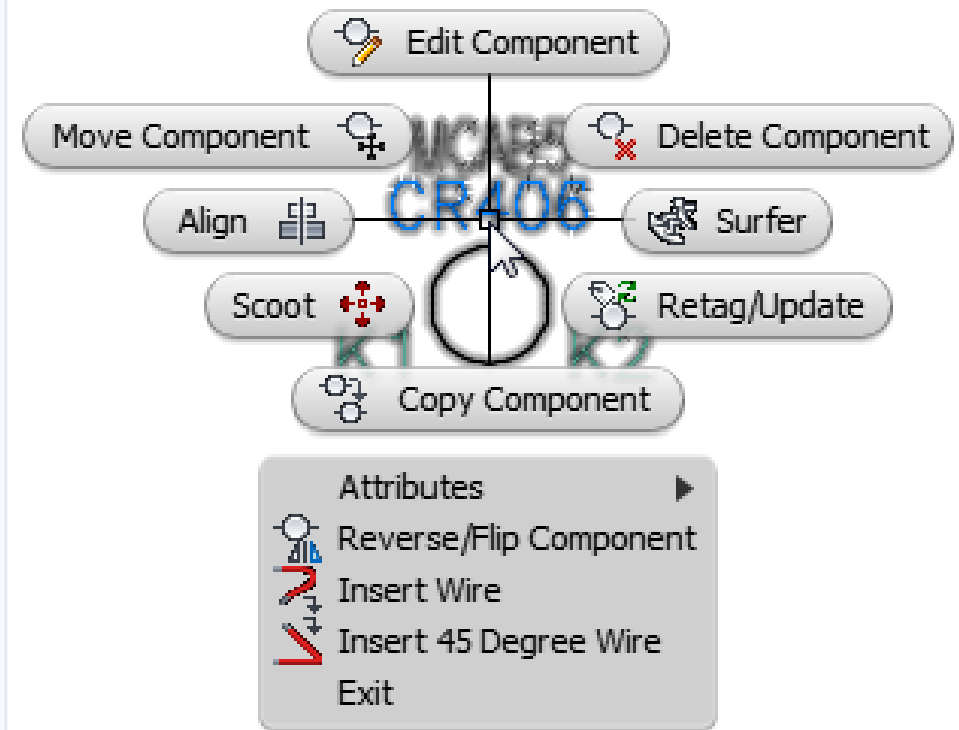
- The Quick Access Toolbar keeps New, Open, Save, Plot and Undo one click away.
- It also carries navigation commands used frequently in AutoCAD Electrical.
- Customise it: click the drop-down at its right end and tick the commands you use most.



Marking Menu

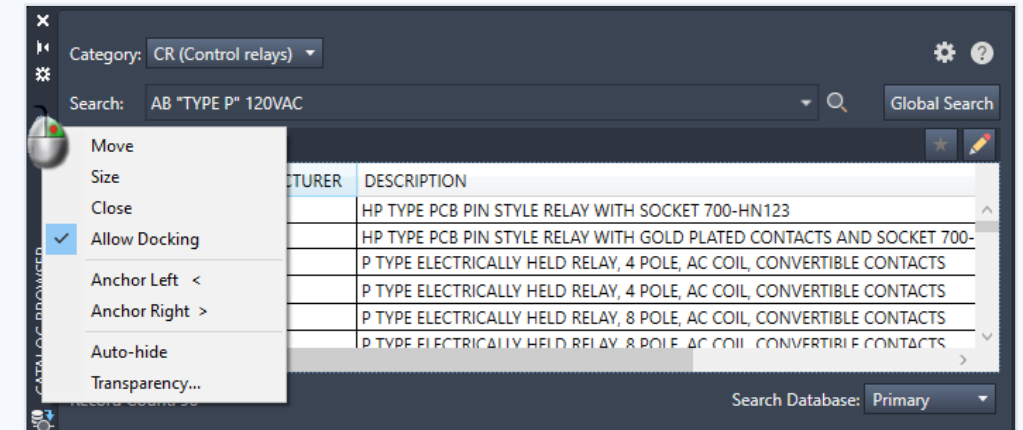
- The marking menu is a radial context menu: hover an electrical object and right-click.
- The options shown depend on the type of object under the cursor — click one to run it.
- Power use: press-and-hold right-click, move the cursor toward the desired option, release — a gesture runs the command directly.

schematic component marking menu



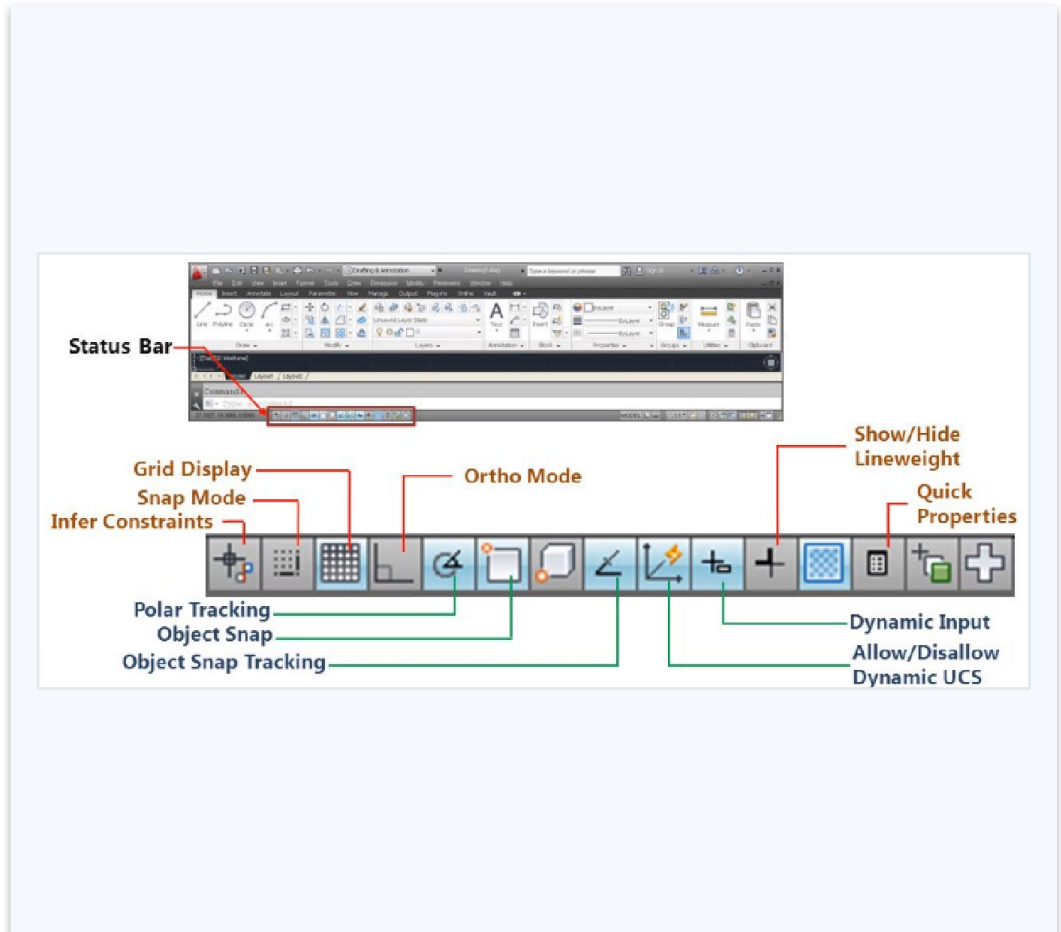
Palettes

- Palettes manage projects (Project Manager) and the catalog database (Catalog Browser).
- A palette stays on screen while other commands run — always accessible.
- Dock, resize or set auto-hide to control how much drawing space a palette uses.



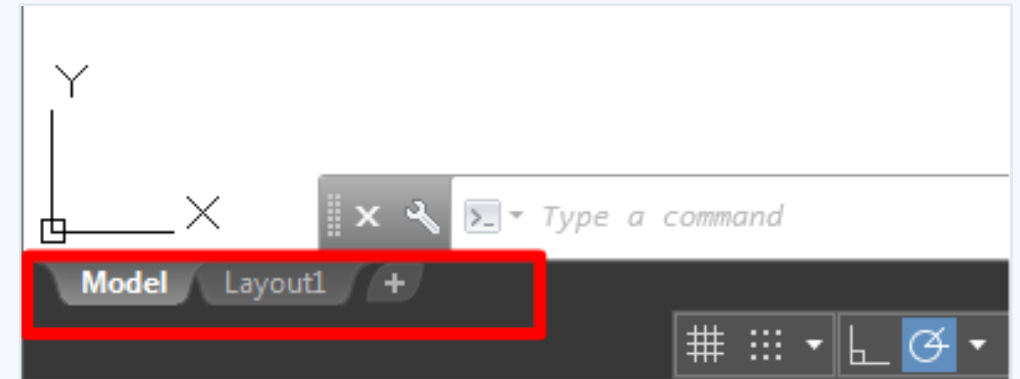
Status Bar — Precision Settings

- The status bar toggles the settings that make a drawing accurate and precise.
- Grid (F7) displays the reference grid; Snap (F9) locks the cursor to grid points.
- Ortho (F8) constrains movement to horizontal/vertical; Polar tracks set angles.
- 2D Object Snap (F3) jumps to endpoints, midpoints and circle centres as you draw.
- These settings are your first production metric: set them before you draw.



Model Space and Paper Space

- Model space is a limitless drawing area where you draw at 1:1 real-world size.
- First decide what one unit represents (1 mm, 1 cm, 1 inch ...) — then draw to that scale.
- Paper space (layouts) arranges scaled views of the model on a titled sheet for plotting.

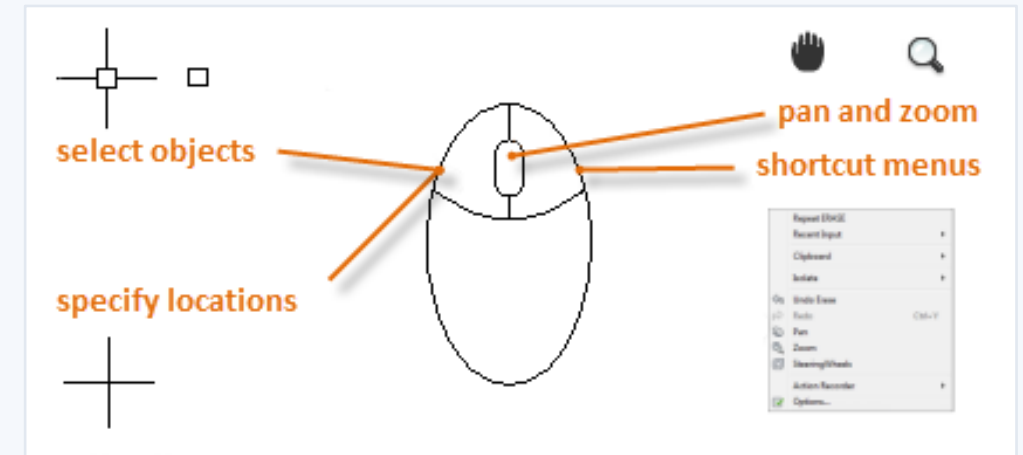


Command Line

- The Command window accepts commands and system variables and prompts you through each command sequence.
- Every ribbon click has a typed equivalent — experienced drafters mix both.
- Ctrl+9 toggles the command window display.
- Watch the prompts: they tell you exactly what input the command expects next.

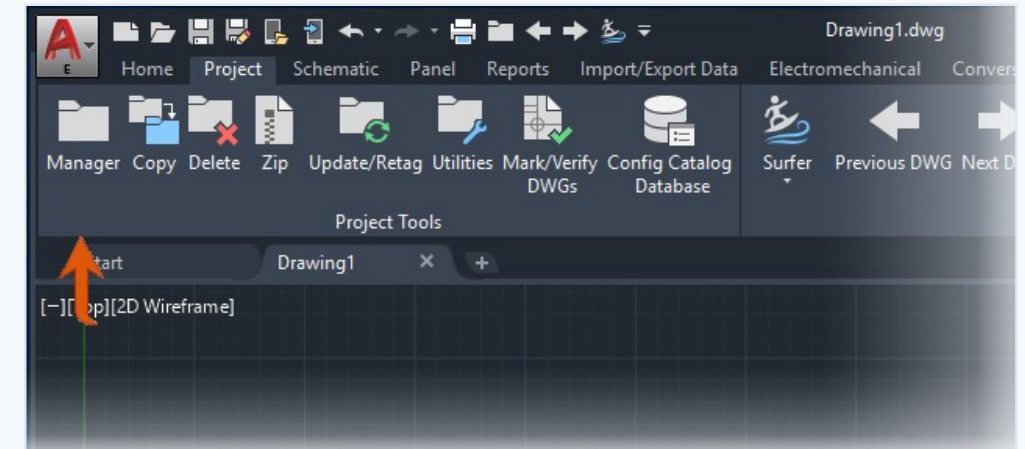
Navigation Tools

- Pan moves the view parallel to the screen; Zoom changes magnification; Orbit rotates 3D views.
- Mouse shortcuts: press the scroll wheel to Pan, roll it to Zoom, double-click it to Zoom Extents.
- ZOOM (Z) options: All shows everything, Window zooms a picked area, Previous restores up to 10 earlier views, Scale magnifies by factor.



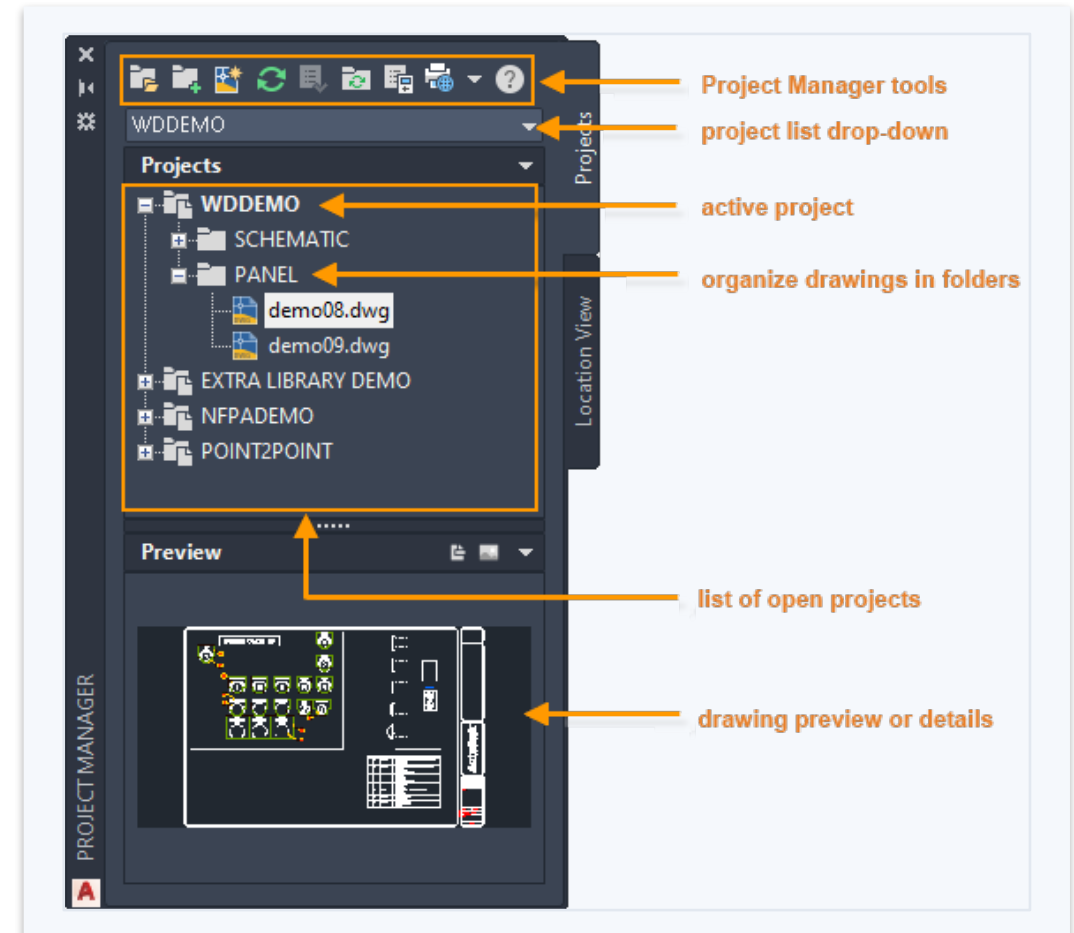
Projects — the .wdp File

- AutoCAD Electrical is project-based: a project is a set of interrelated drawings.
- All project information — including the drawing list — is saved in a .wdp project file.
- Project Manager creates, opens and manages projects and their settings.
- Working project-wide is what lets tags, wire numbers and cross-references stay consistent.



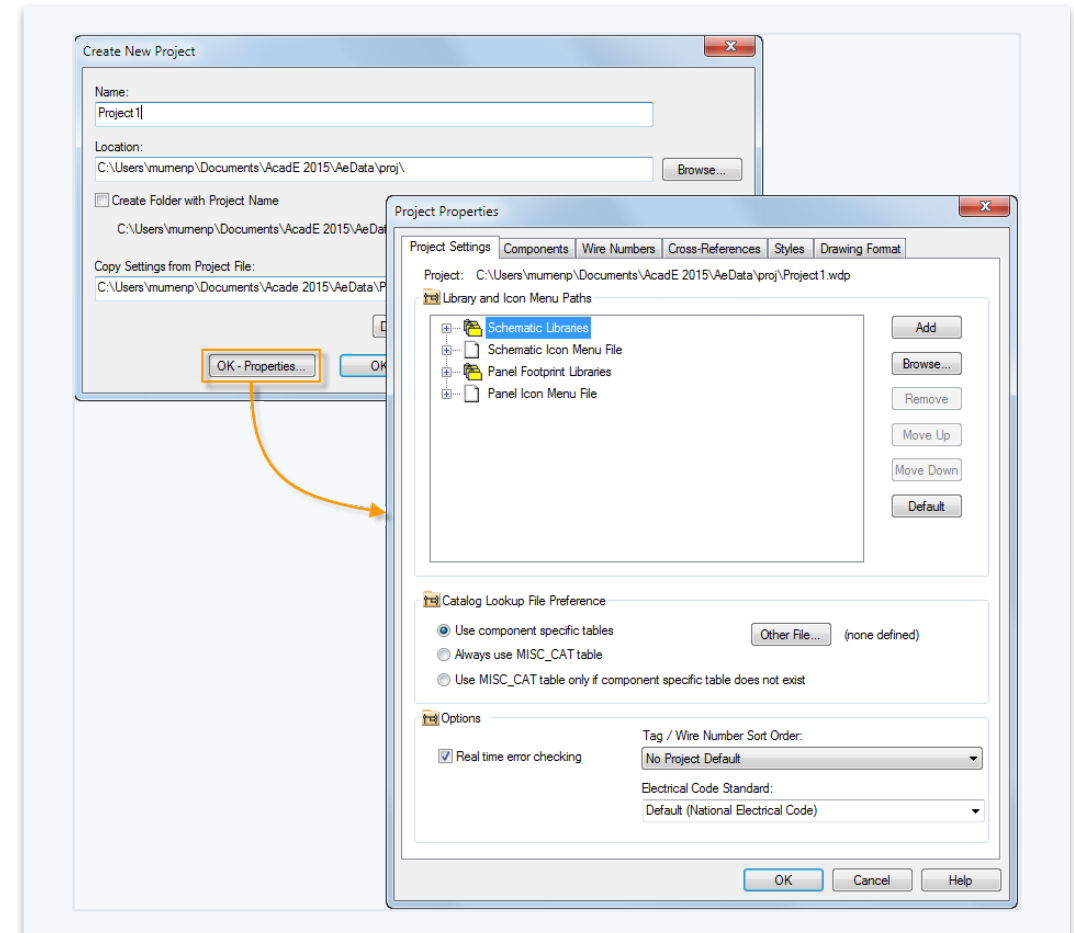
Project Manager

- Project Manager is a palette — it can stay on screen while you use other commands.
- It can be docked, resized and set to auto-hide.
- Create a project and the .wdp file lands in your chosen folder; the new project becomes active and you add drawings to it.



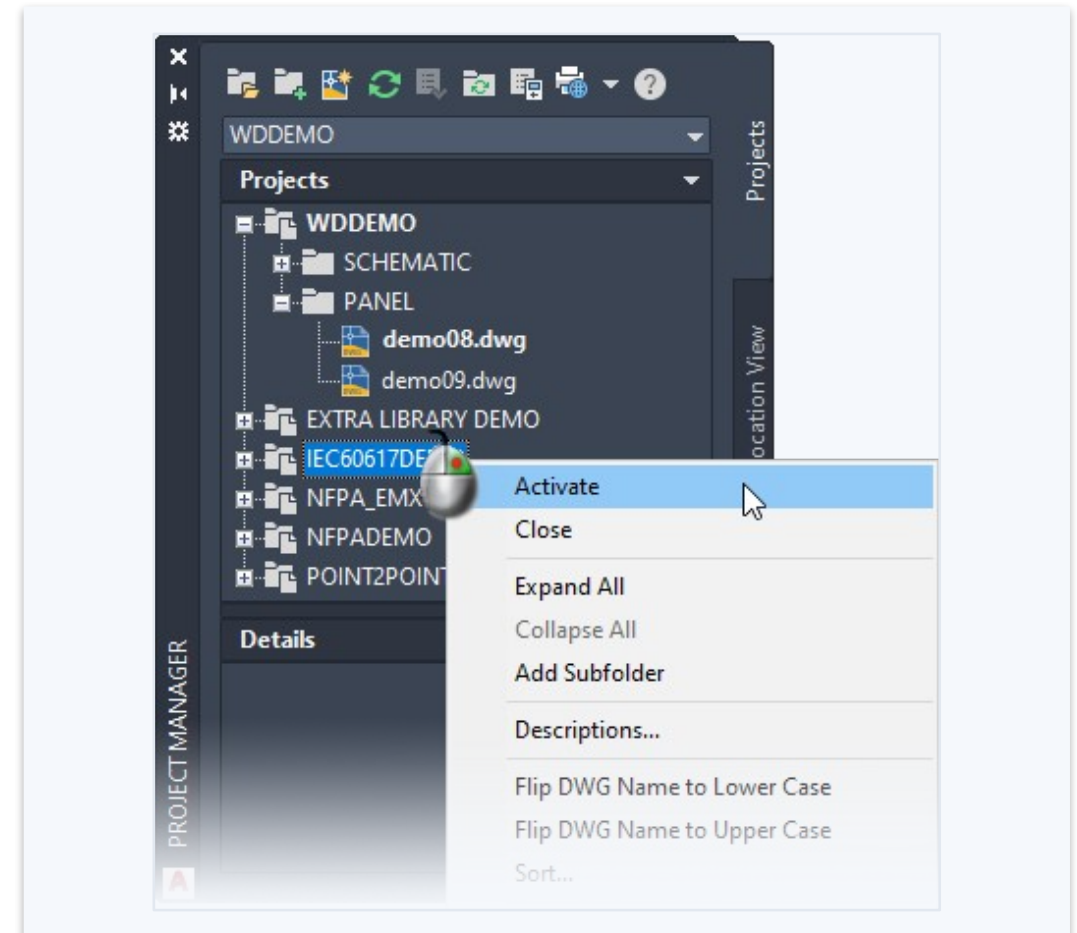
Project Properties

- Define library paths, drawing format and reference style when you start a project — or any time later.
- Two kinds of properties: drawing defaults (copied into each new drawing) and project-wide settings.
- Setting properties once at project level is how an organisation standardises its drawings.



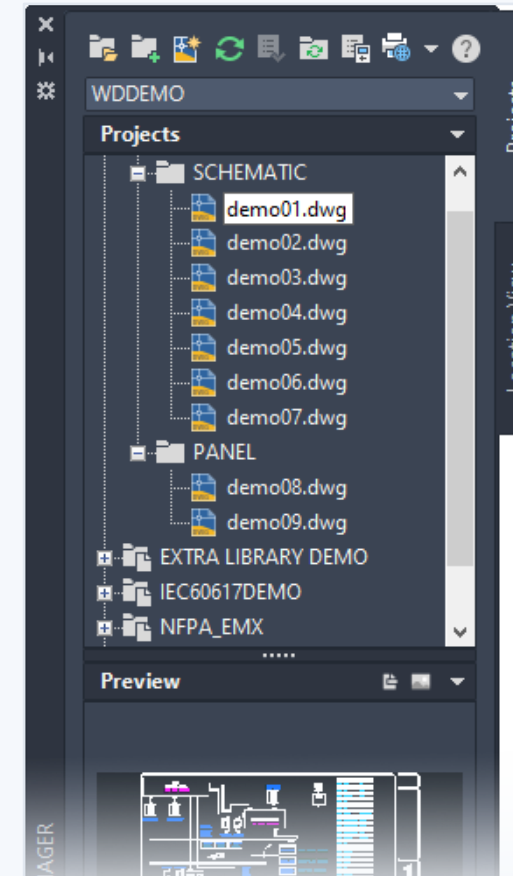
Activate a Project

- Activate from the list of open projects in Project Manager.
- Or browse to the project's .wdp file, or pick from the recent-projects list.
- An opened project joins the Project Manager list and becomes the active project.
- Copy an existing project to reuse its properties, drawings and support files in a new job.



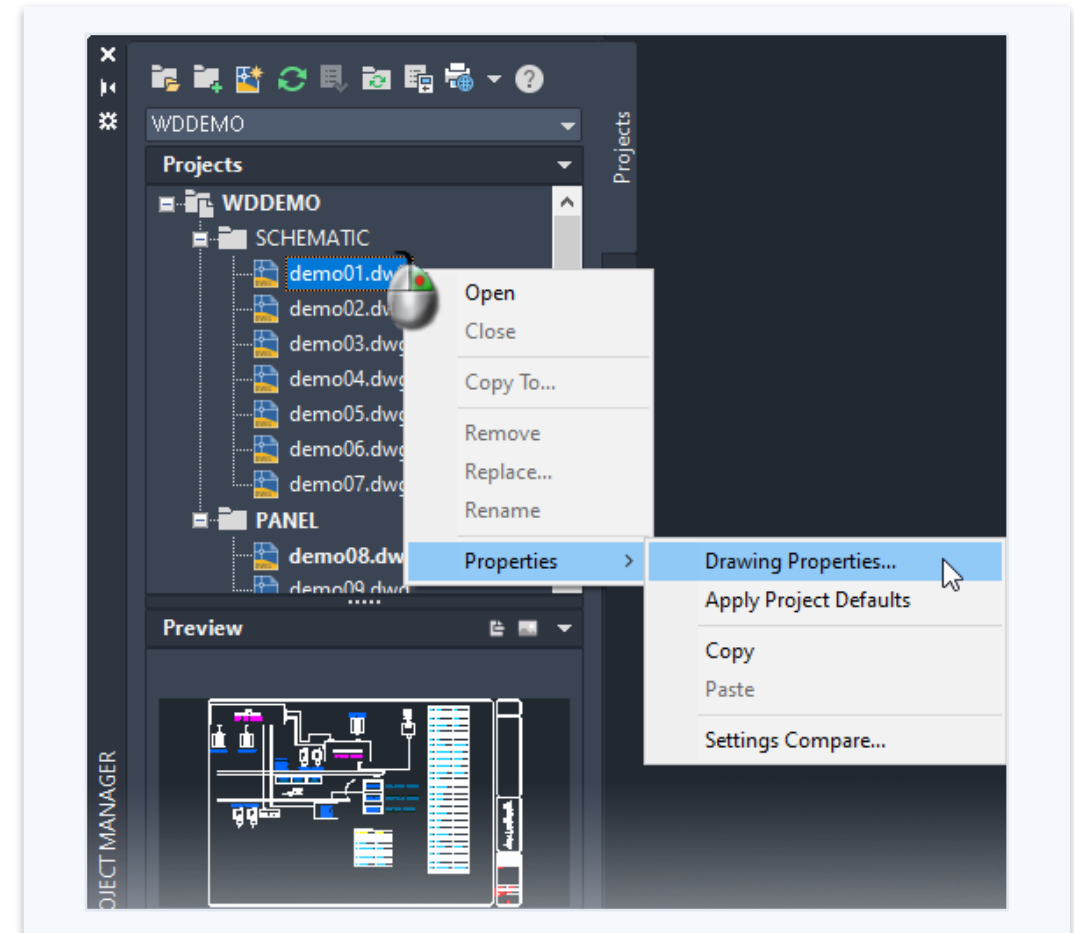
Drawings in a Project

- The drawing order sets the processing order for project-wide functions — drag and drop to reorder.
- Add folders within a project to organise drawings.
- New Drawing creates a drawing inside the active project; give it a template with predefined electrical properties, wire layers, a reference system and a title block.



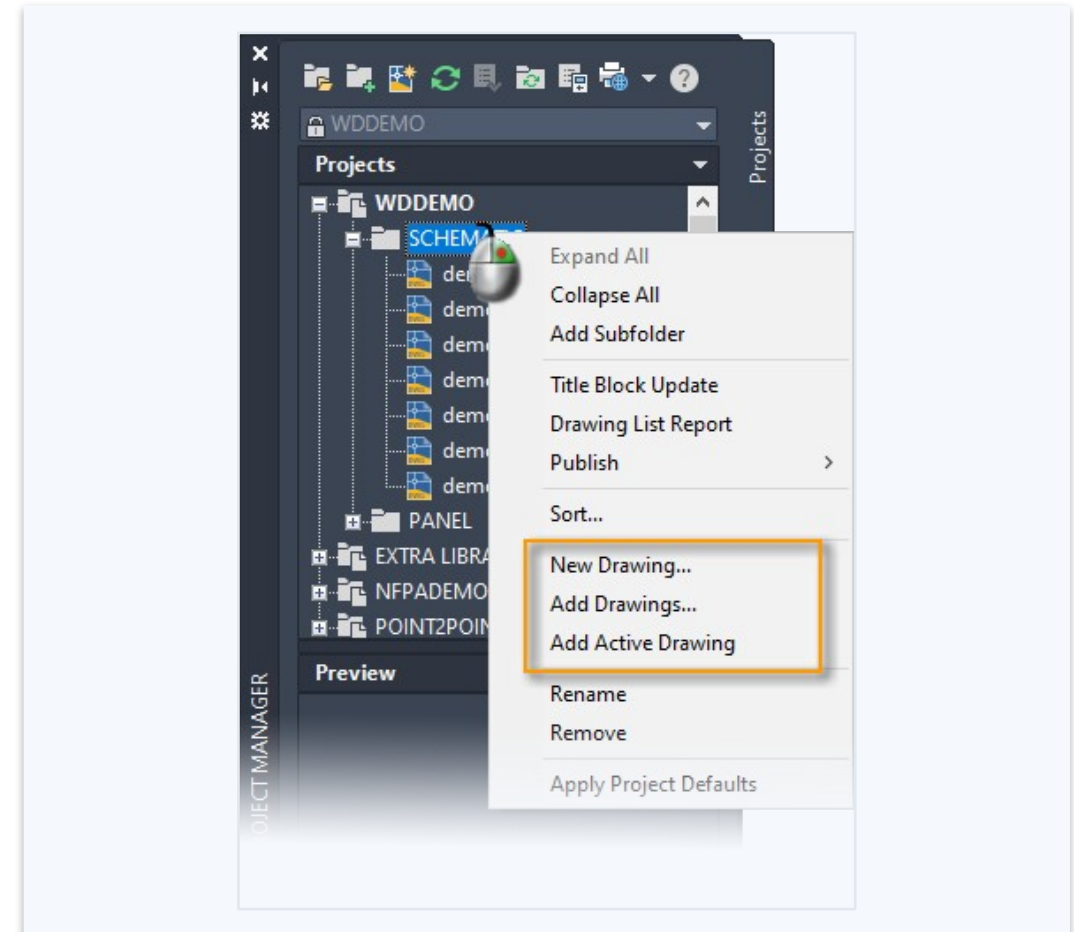
Drawing Properties

- Drawing Properties control tagging and referencing per sheet: sheet values, installation and location codes.
- Reference system: ladder, X-Y Grid or X Zones; formats for component tags, wire numbers and cross-references.
- Styles: PLC modules, wire crossing and wire connection appearance.
- Properties come from the template or the project defaults — override via OK-Properties when creating the drawing.



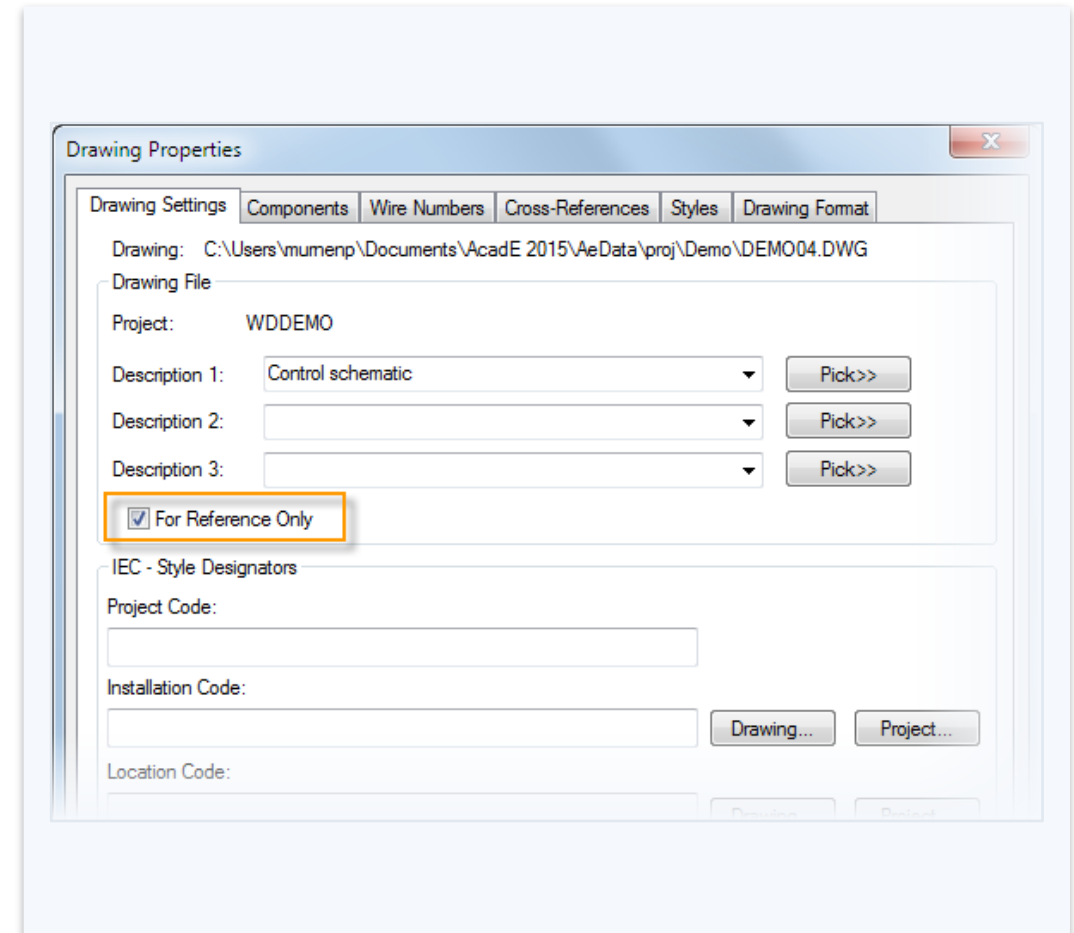
Copy and Add Drawings

- Copy a single drawing from any open project into the active project via Project Manager.
- Add existing drawings any time — the way to reuse sheets from a previous job.
- After adding, run Project-Wide Update/Retag to refresh ladder references, component tags, wire numbers and cross-references.



Reference-Only Drawings

- Some drawings belong to the project but must not be processed by project-wide functions.
- Mark them Reference Only in the drawing properties — tagging, cross-referencing and reporting skip them.
- Typical examples: cover sheets, general notes, mechanical outlines.

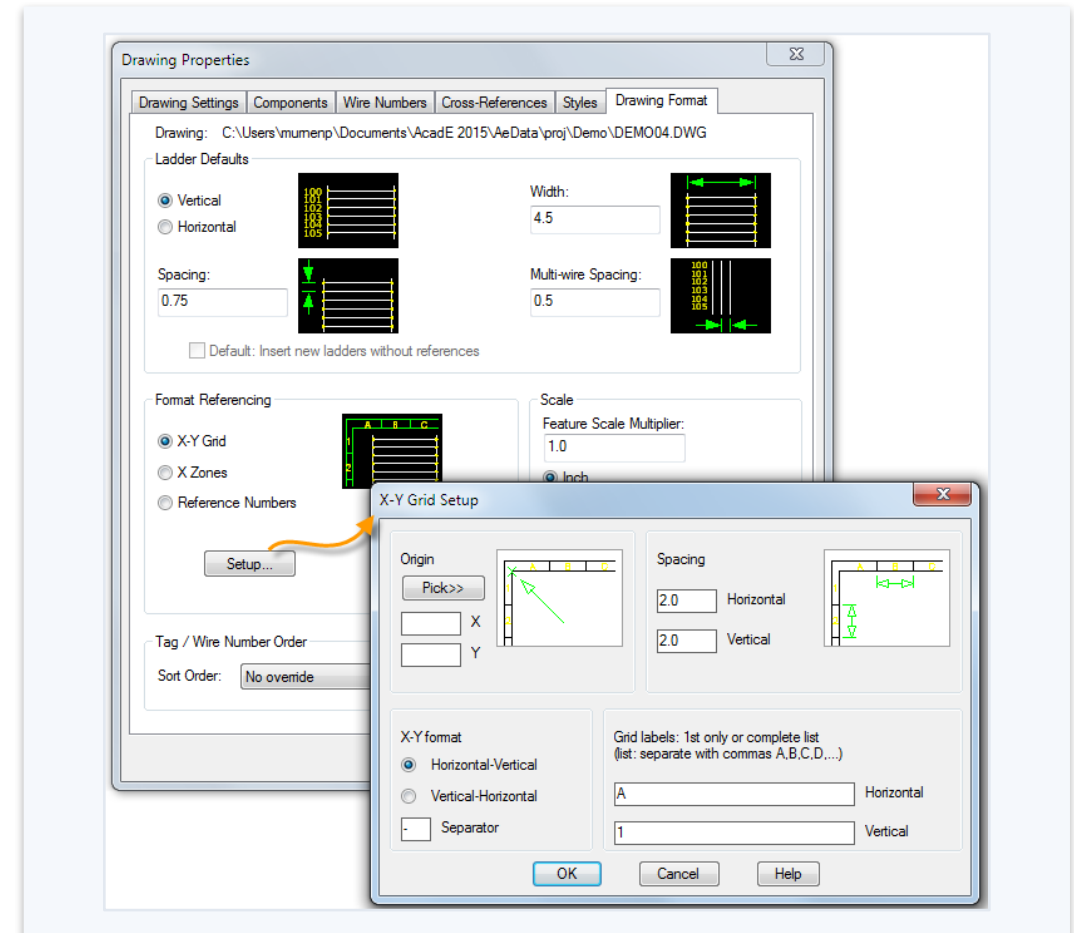


Title Block Setup

- Insert a .dwg as the title block and link project/drawing property values to its attributes.
- Title Block Setup creates and defines the mapping between properties and attributes.
- Once mapped, sheet number, description and revision fill in automatically on every sheet — a core drafting standard.

Reference Systems — Ladders, X-Y Grid, X Zones

- Each schematic drawing can define a reference system: ladder rungs, X-Y grid or X zones.
- The reference system assigns unique component tags and wire numbers (unless sequential tagging is used).
- It also drives cross-referencing between parents and children.
- Insert the ladder or grid from Schematic → Insert Wires/Wire Numbers; define it on a template to save time.



Hands-On Labs — Topic 1

Lab 1

- Set Up a New AutoCAD Electrical Drawing
- A correctly configured metric drawing sheet, saved as a reusable baseline for the rest of the course.

Lab 2

- Create a Project with Project Manager
- A working .wdp project containing the supplied power and lighting plan drawings plus a title block, in the correct sheet order.

Set Up a New AutoCAD Electrical Drawing

LAB 1

Create a new drawing from the ACADISO template, then set the workspace, drawing limits and units — the production metrics every drawing in your organisation should start from.

You'll build

A correctly configured metric drawing sheet, saved as a reusable baseline for the rest of the course.

Tools: AutoCAD Electrical · ACADISO template · Status bar · Grid & Snap

Set Up a New AutoCAD Electrical Drawing

1

Start AutoCAD Electrical and create a new drawing from the ACADISO template (New → acadiso.dwt).

NEW → acadiso.dwt

2

Set the Electrical workspace from the Workspace Switching control on the status bar.

WSCURRENT → Electrical

3

Set the drawing limits to an A3 metric sheet (0,0 to 420,297) and Zoom All to see the full sheet.

LIMITS 0,0 420,297 then ZOOM All

4

Set the units to decimal millimetres with 0 precision (Format → Units).

UNITS → Decimal · 0

Set Up a New AutoCAD Electrical Drawing

5

Turn on Grid (F7) and Snap (F9) from the status bar and open DraftingSettings.dwg to compare a pre-configured sheet.

`GRID 10 · SNAP 10`

6

Save the drawing as MyBaseline.dwg — this is your production guideline drawing.

`SAVEAS MyBaseline.dwg`

Set Up a New AutoCAD Electrical Drawing

Test it

ZOOM All shows the full 420×297 sheet, the crosshair snaps in 10 mm increments, and STATUS reports decimal units — your baseline metrics are in force.

Create a Project with Project Manager

LAB 2

AutoCAD Electrical is project-based: a .wdp project groups interrelated drawings so project-wide functions can retag and renumber. Create a project, add the supplied electrical plan drawings, and reorder them.

You'll build

A working .wdp project containing the supplied power and lighting plan drawings plus a title block, in the correct sheet order.

Tools: Project Manager · .wdp project file · Power_Plan / Lighting_Plan DWGs

Create a Project with Project Manager

1

Open Project Manager from the Project tab (it is a palette — dock it on the left).

AEPROJECT

2

Create a new project named TRAINING — the .wdp file stores the project description and settings.

Project Manager → New Project →
TRAINING

3

Review the Project Properties: library standard (IEC/IEEE/NFPA), component tag format and cross-reference style.

Right-click project → Properties

4

Add the supplied drawings 10-2_Power_Plan.dwg and 10-2_Lighting_Plan.dwg to the project.

Right-click project → Add Drawings

Create a Project with Project Manager

5

Drag and drop the drawings inside Project Manager so the power plan is sheet 1, and add Title_Block.dwg as reference-only.

Drag to reorder · Properties → Reference Only

6

Make TRAINING the active project and open a drawing from the project list by double-clicking it.

Right-click → Activate

Create a Project with Project Manager

Test it

Project Manager shows **TRAINING** in bold (active) with the drawings in your chosen order, and the .wdp file exists in the project folder — project-wide functions now see every sheet.

Recap — Topic 1

- A1 — devise metrics and guidelines for technical drawing production (template, workspace, limits, units) (Lab 1)
- A1, A2 — standardise production with a project; solve multi-drawing organisation problems (Lab 2)
- Key concepts: Symbols, standards, conventions, CAD vs manual drawing, The AutoCAD Electrical UI, Projects drive everything, Devise metrics & guidelines, Formulate solutions.

Lunch Break

Back at 1245 · PM digital attendance

TOPIC 02

Analysing Technical Drawings

Types of schematic drawings · components, terminals & PLC symbols · wires & part lists · accuracy review (K3, K4 · A3, A4)

Key Concepts — Topic 2

1

Types of technical drawings

Schematics, panel layouts, wiring diagrams, one-line diagrams and reference-only sheets each serve a construction purpose. (K3)

2

Technical drawing standards

Ladder references, X-Y grids, wire-number formats and cross-reference styles are the standards drawings are checked against. (K4)

3

Schematic components

Insert from the icon menu or catalog database; annotate with tags, descriptions and catalog data.

4

Terminals & PLC symbols

Standard part lists rely on terminal strips and parametric PLC I/O modules that adapt to the ladder.

5

Wires, numbers & arrows

Wire layers, wire numbers and source/destination signal arrows carry the circuit across sheets.

6

Review for accuracy

Cross-referencing, Surfer and Mark/Verify audit a drawing set for accuracy and adherence to specifications. (A3, A4)

Types of Technical Drawings

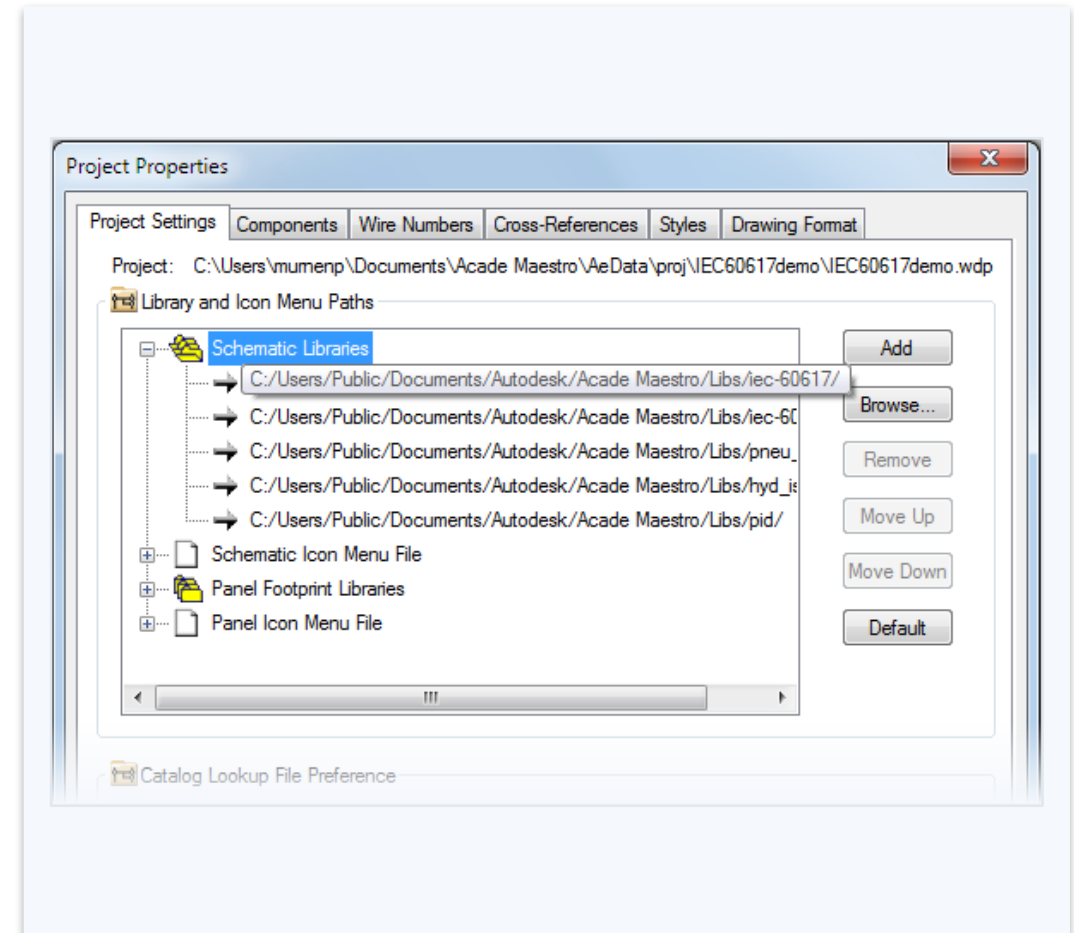
- Schematic diagrams show circuit logic: ladders, components, wires and signals.
- Panel layout drawings show the physical arrangement of components in an enclosure.
- Wiring / connection diagrams document point-to-point conductor runs and terminal strips.
- One-line (single-line) diagrams summarise power distribution at system level.
- Reference-only sheets (covers, notes, outlines) support the set without carrying circuit data.
- Each type answers a different construction need — a full drawing set uses them together.

Schematic Components — Grid and Snap Discipline

- Insert schematic components, generate PLC modules, and insert or copy circuits.
- Work with grid lines on and snap-to-grid when inserting components.
- GRID sets grid spacing; SNAP sets snap spacing; F7 toggles grid visibility, F9 toggles snap.
- Consistent snap spacing keeps rungs aligned so wires connect cleanly — an accuracy standard.

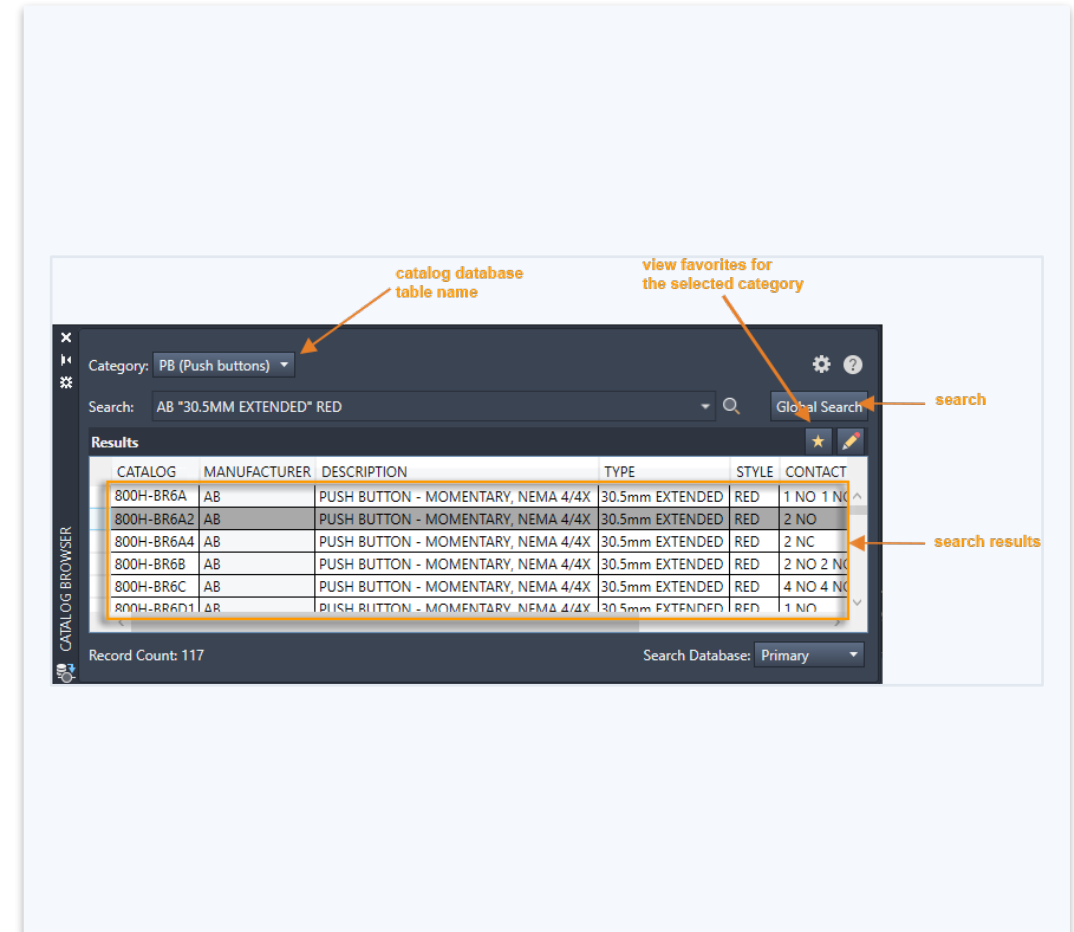
Symbol Library Standards

- AutoCAD Electrical ships symbol libraries complying with IEEE 315/315A, IEC-60617 and NFPA.
- IEEE and IEC libraries are metric; NFPA library symbols are in inches (scale with the Modify-library utility).
- The old JIC standard was absorbed into NFPA 79; JIC and older IEC libraries remain for legacy support.
- Pick ONE standard per project in Project Properties — mixing standards breaks readability across regions.



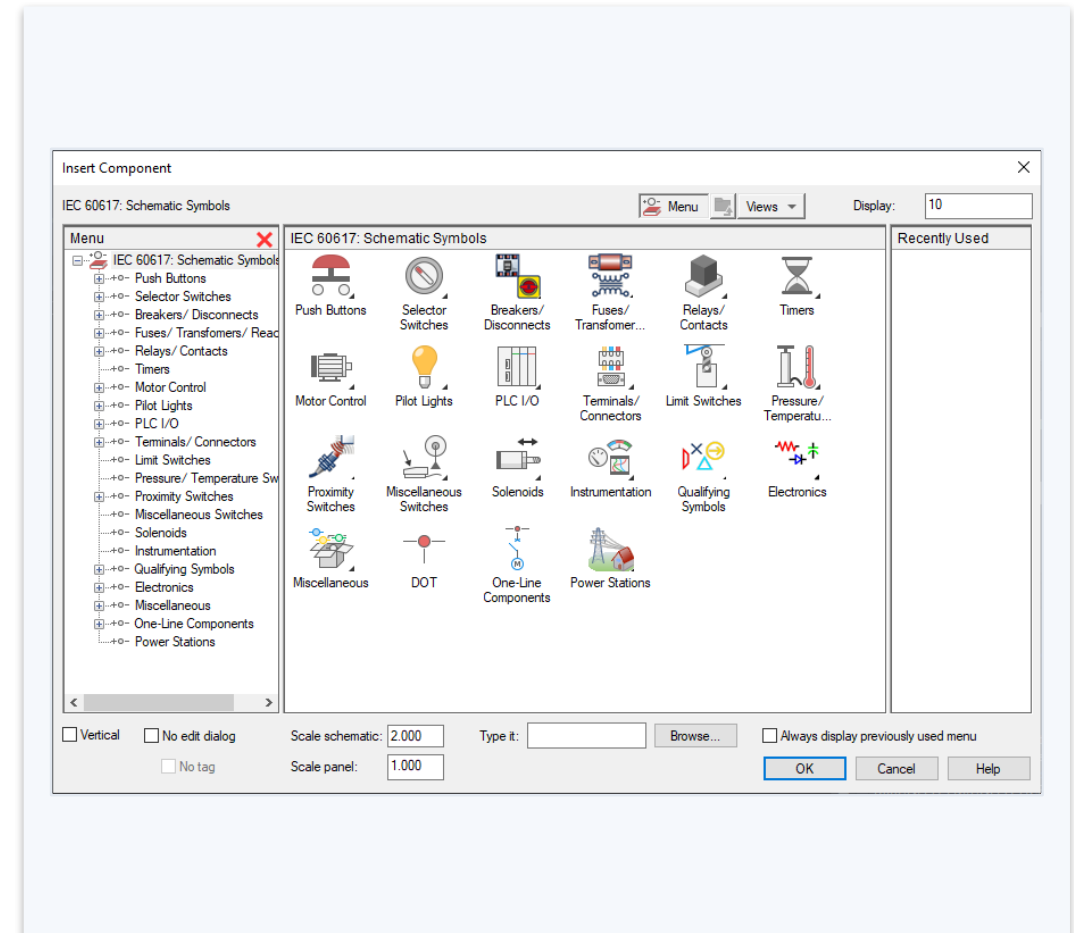
Insert a Component from the Catalog Database

- Know the part number already? Insert the component straight from the catalog database.
- Use the Catalog Browser palette; search values are saved for quick reuse.
- Catalog information lands on the symbol automatically; edit further in the Insert/Edit dialog that follows insertion.



Insert a Component from the Icon Menu

- The icon menu inserts by symbol type — a NO push button, a 3-position selector switch, and so on.
- Different icon menus exist per library set; choose the project's icon menu under Project Settings.
- After insertion, annotate with descriptions and catalog information.



Annotate the Symbol

- Each symbol carries a main catalog value plus optional multiple-catalog values.
- Edit a component to add, change or remove catalog assignments — Lookup opens the catalog search.
- Multiple Catalog attaches auxiliary parts (contact blocks, adapters) to the same device.
- Complete, accurate annotation is what makes reports and BOMs trustworthy downstream.

Insert / Edit Component

Component Tag
 CR406 ☐ fixed
 Use PLC Address
 Tags Used: Schematic Panel External List Options

Description
 Line 1: SAFETY
 Line 2: RELAY NO.1
 Line 3:
 List: Drawing Project Defaults Pick

Catalog Data
 Manufacturer: AB
 Catalog: 700-P200A1
 Assembly:
 Item: Count:
 Next>> 1
 Lookup Previous
 Drawing Project
 Multiple Catalog
 Catalog Check

Cross-Reference
☐ Component override Setup
 Reference NO: NO/NC Setup
 Reference NC:

Installation code
 Drawing Project

Location code
 MCAB5
 Drawing Project
 Show/Edit Miscellaneous

Pins

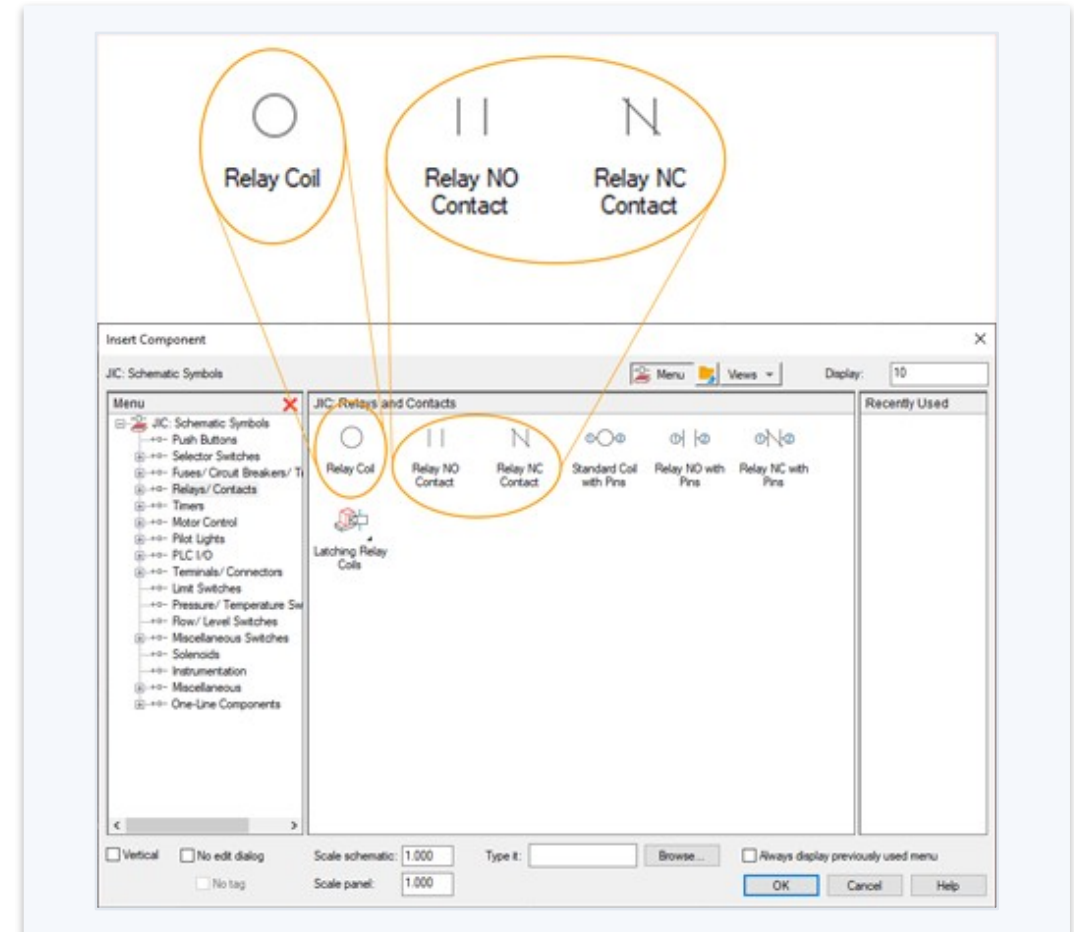
Pin	Terminal	Direction
1	K1	< >
2	K2	< >
×		< >

Rating: Show All Ratings

OK-Repeat OK Cancel Help

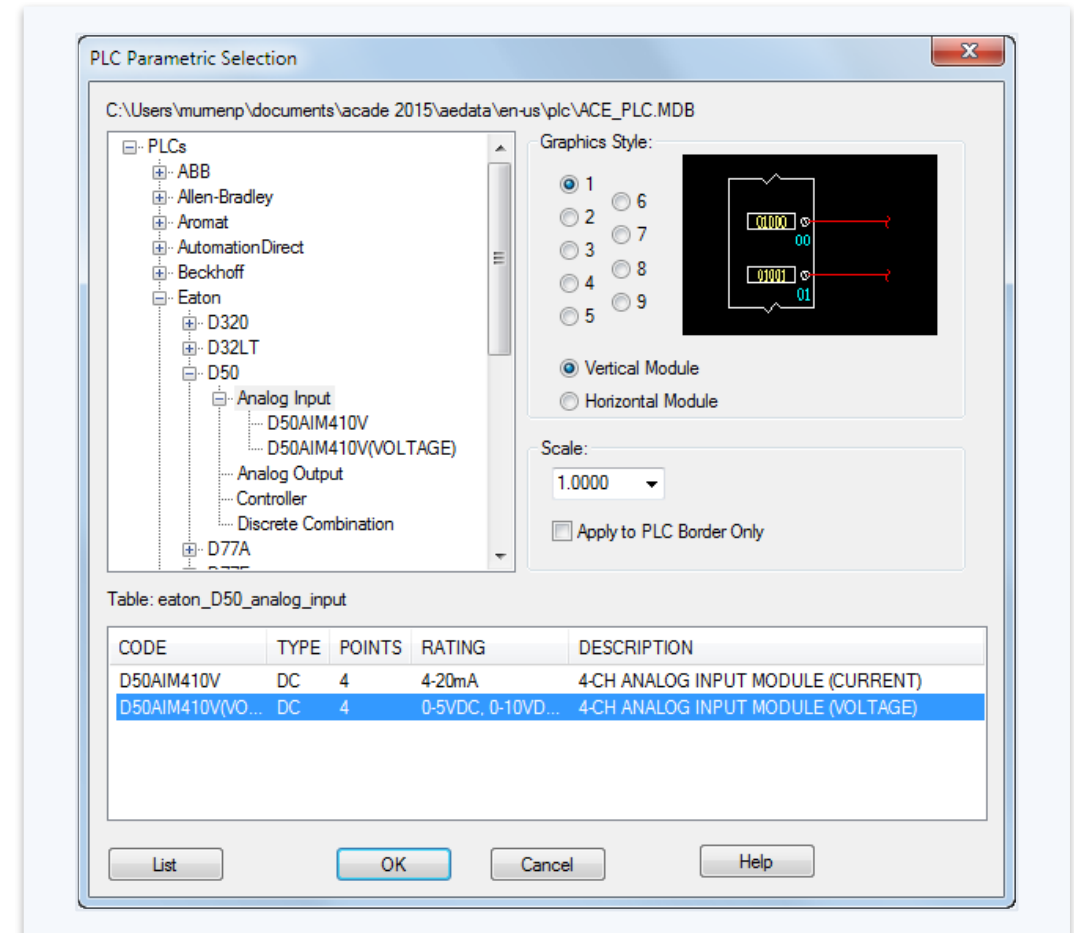
Parent and Child Components

- Some devices are drawn as several symbols — a relay is a coil plus its contact symbols.
- The coil is the parent; contacts are children that inherit its tag and data.
- When inserting a child, pick its parent from the active drawing or project list in the Insert/Edit Child dialog.
- Parent/child links power the cross-reference grid you will read in every review.



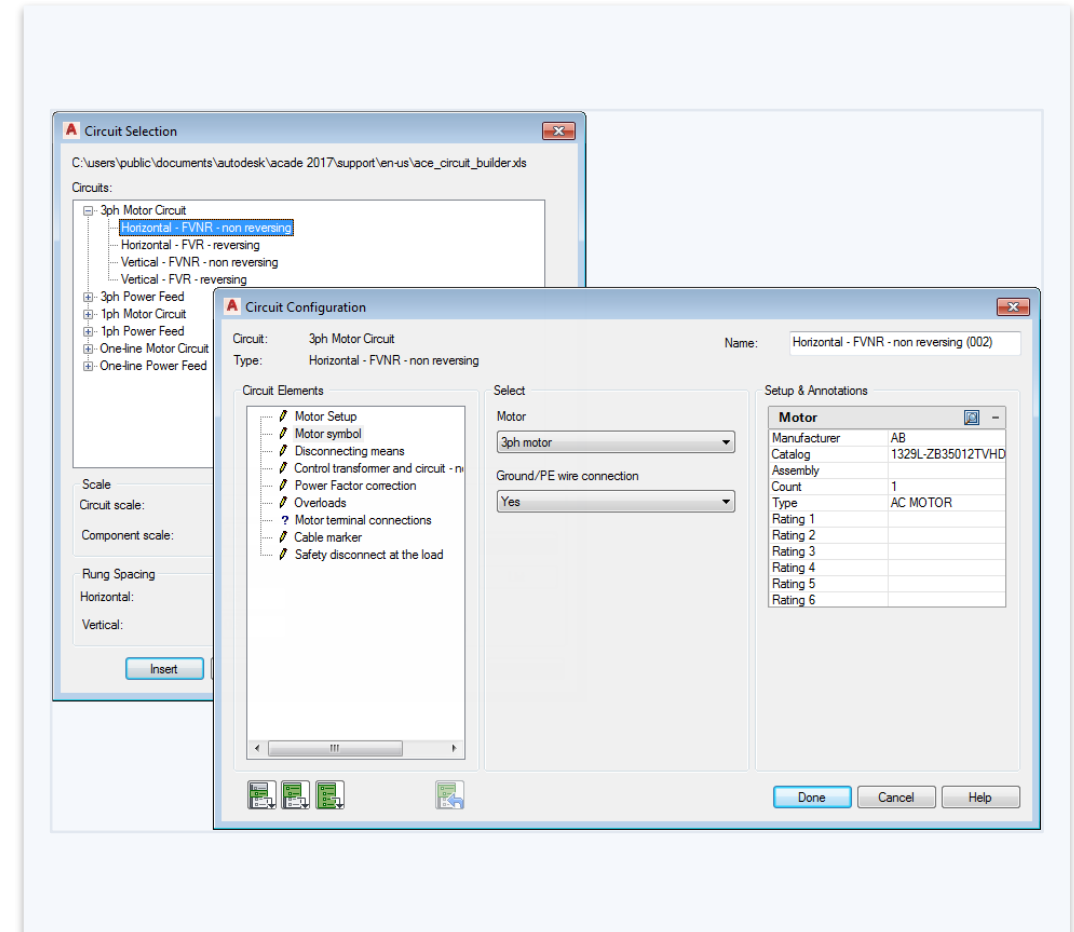
PLC I/O Modules

- AutoCAD Electrical generates hundreds of PLC I/O modules on demand, in multiple graphical styles.
- Modules adapt automatically to the underlying ladder rung spacing.
- Expand manufacturer → series → type, pick the module and style, then place it by insertion point.
- Parametric generation keeps PLC drawings consistent with the wiring standard in force.



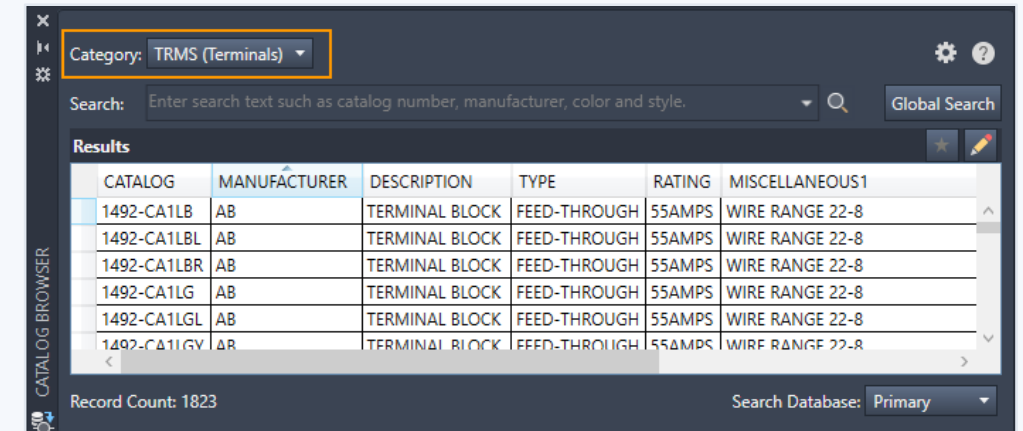
Circuits and Circuit Builder

- A circuit is any collection of components and wires; save frequently used circuits to the icon menu.
- A saved circuit re-inserts with tags updating automatically to stay unique.
- Circuit Builder assembles motor-control and power-feed circuits dynamically from your dialog selections.
- Building from proven circuits is how construction needs are met quickly and correctly.



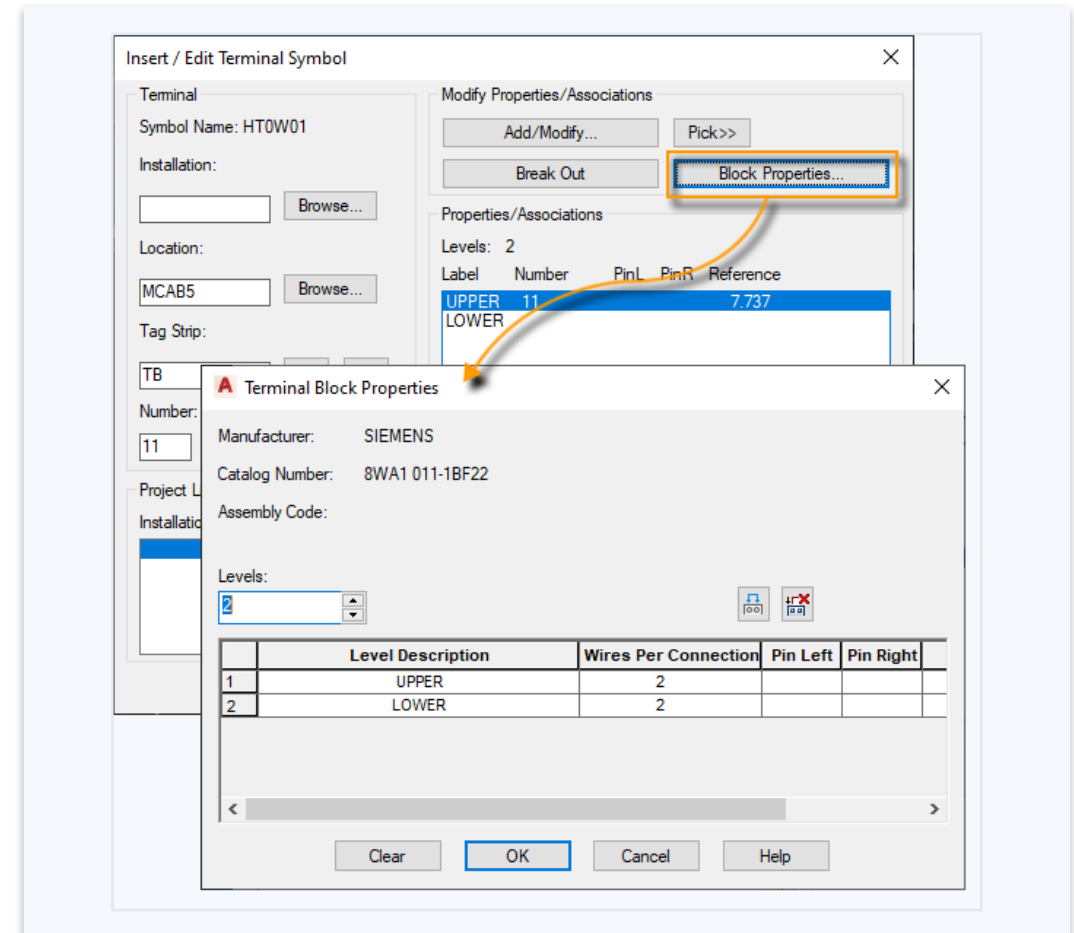
Schematic Terminals

- Insert terminals like other schematic symbols — from the Catalog Browser (TRMS category) or the icon menu.
- Four terminal behaviour types exist; the terminal block name controls the behaviour.
- Pick the behaviour type and terminal style carefully when inserting from the icon menu.
- The TERMINAL STRIP command inserts strip symbols using Installation, Location and Tagstrip values.



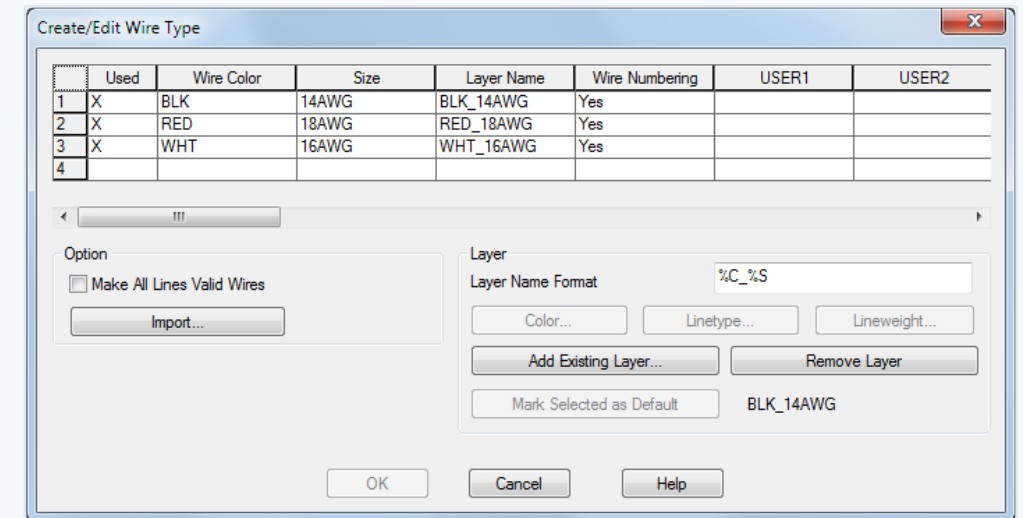
Terminal Block Properties

- A schematic terminal symbol represents a single-level terminal or ONE level of a multi-level terminal.
- The number of levels is a block property; each level carries its own wiring data.
- View and edit properties in the Insert/Edit Terminal Symbol dialog.
- Associate schematic terminals across levels — and manage whole strips in the Terminal Strip Editor.



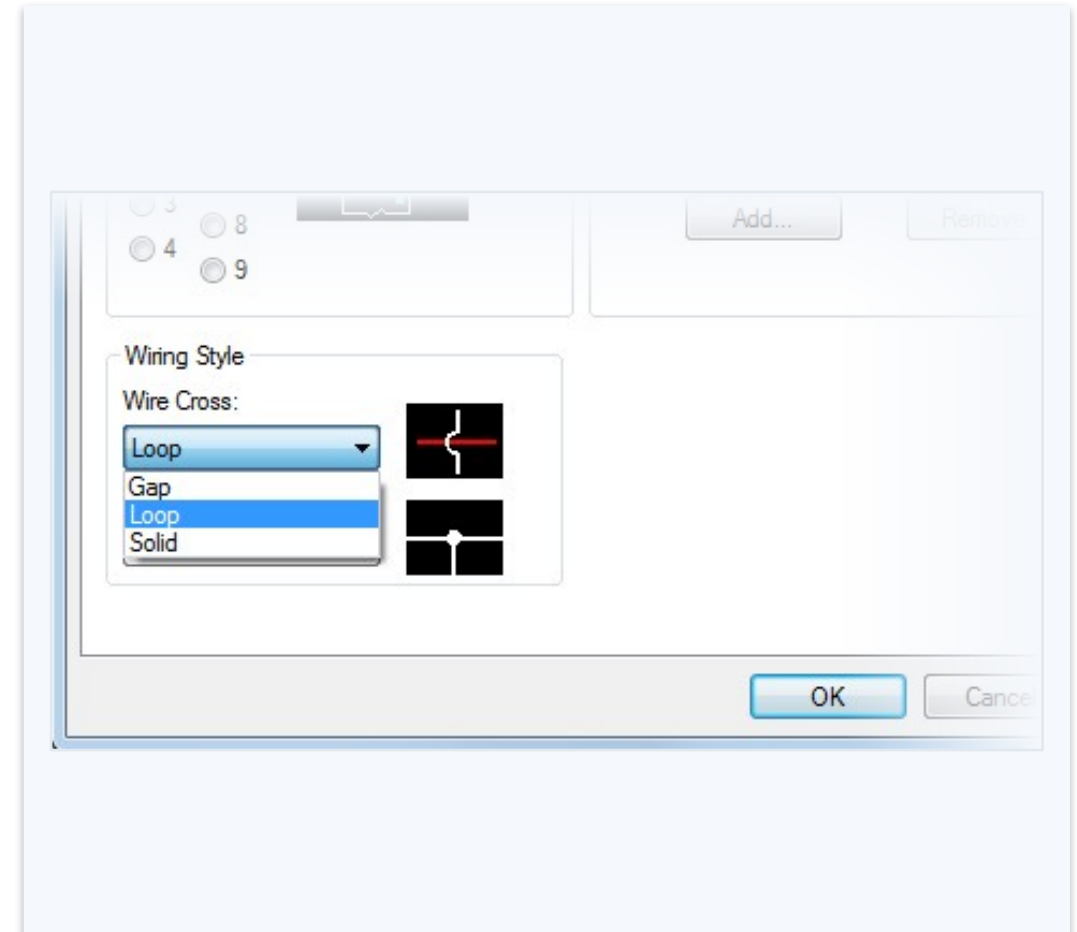
Wires and Wire Layers

- A wire is a line object on a layer defined as a wire layer; a wire network is connected wire segments.
- Manage wire layers in a grid: edit colour and size inline, right-click to rename.
- Create layers for wire families (24VDC, 230VAC, earth) so wiring intent is visible at a glance.



Insert Wires

- Inserted wires land on a wire layer automatically — joining an existing wire adopts that wire's layer.
- A tee connection receives a dot or angled tee per the Styles tab of Drawing Properties.
- Change/Convert Wire Type moves wires to another layer or converts plain lines into wires.

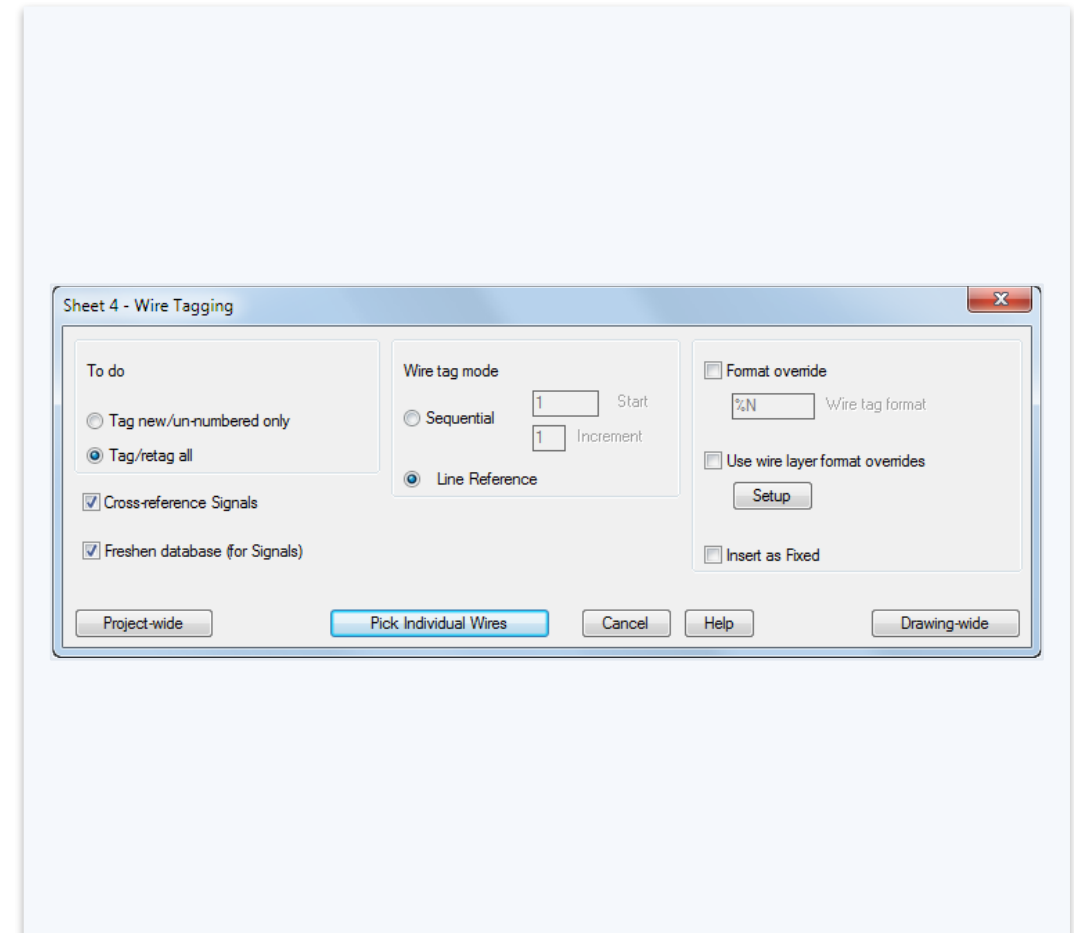


Wire Number Setup

- The Wire Numbers tab of Drawing Properties defines the wire-number format, type and placement.
- The format mixes text and replaceable parameters: %N (number), %S (sheet), %D (drawing), %X (suffix position), %P/%I/%L (IEC project / installation / location codes).
- Wire numbering is a specification — every sheet in the set must follow the same format.

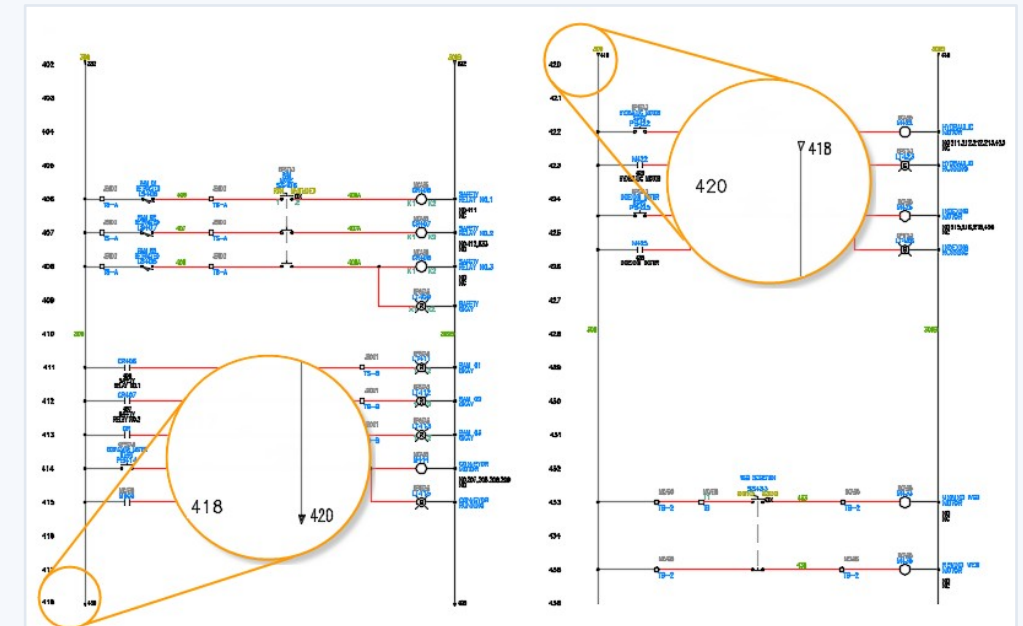
Insert Wire Numbers

- The Wire Numbers command inserts new numbers or updates existing ones drawing- or project-wide.
- Drawing-property settings are preselected; override per run if needed.
- Numbers marked Fixed are preserved by renumbering — edit them via Edit Wire Number.



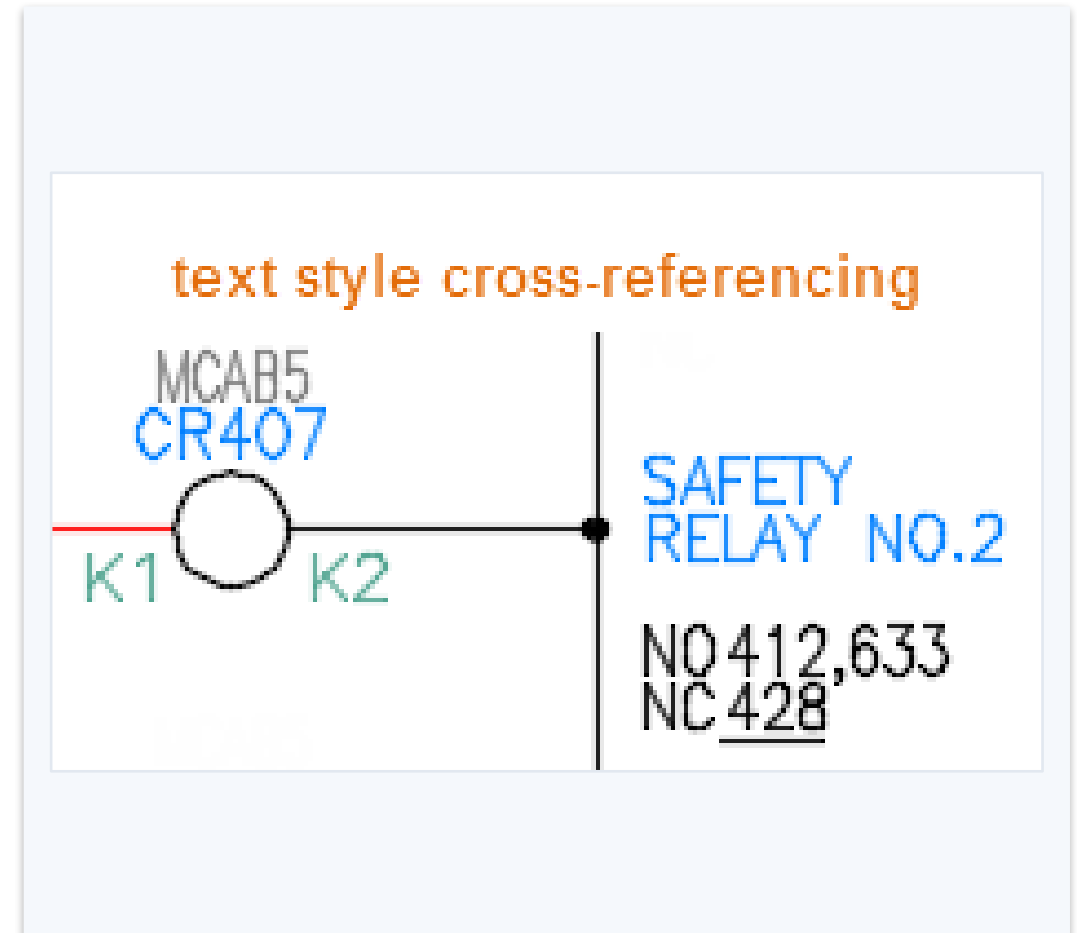
Signal Arrows

- A signal arrow breaks a wire network and continues it elsewhere — same drawing or another sheet.
- AutoCAD Electrical uses named source/destination pairs: give the source a code, then insert destinations that reference it.
- Arrows carry cross-reference annotation both ways, so a checker can trace the signal across the set.



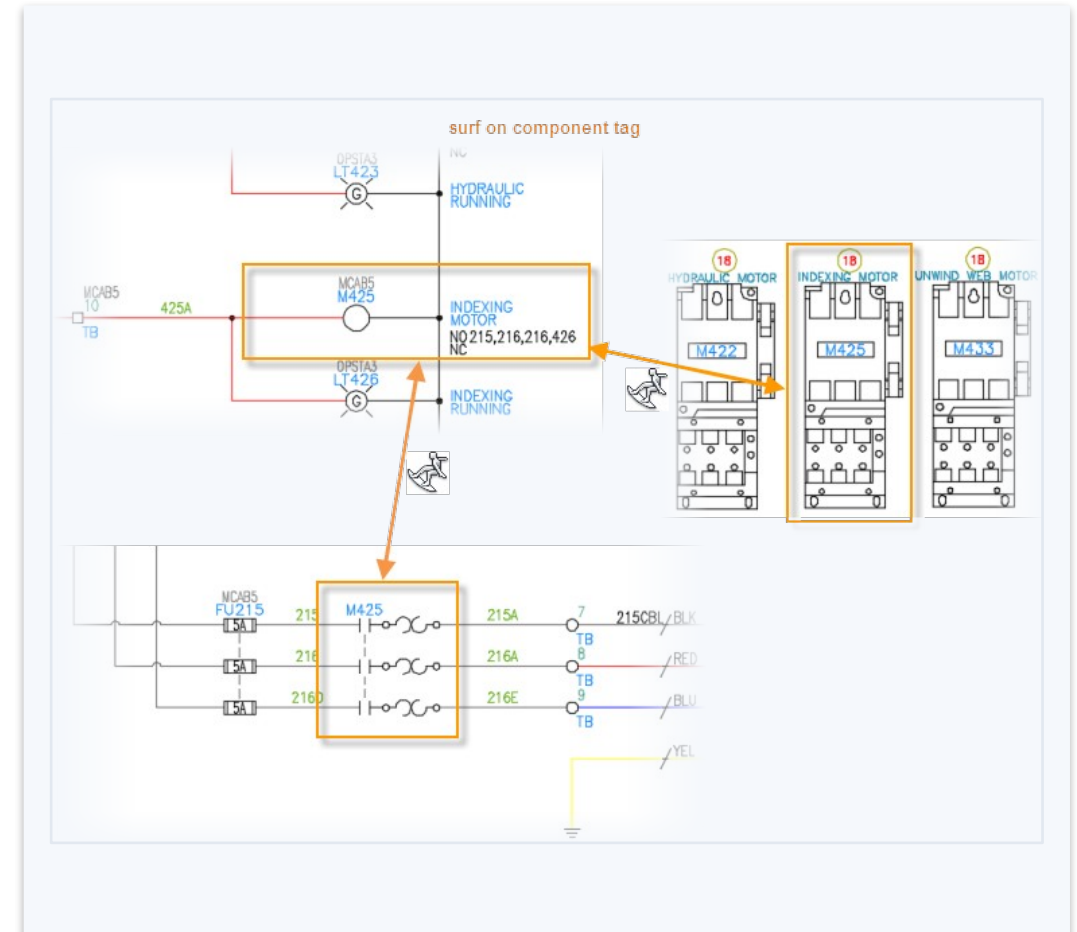
Cross-Referencing

- Cross-references annotate related components with each other's reference locations.
- Three annotation styles on a parent; child annotation is always text format.
- Signal arrows are cross-referenced too — source shows destinations, destination shows its source.
- Set the format and style on Drawing Properties → Cross-References; keep it current before any review or PDF publish.



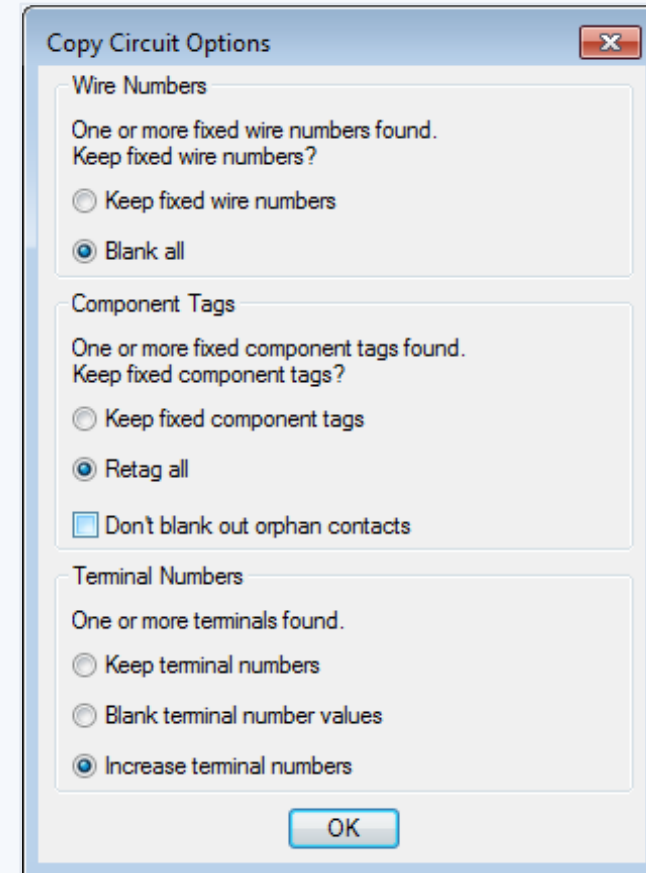
Surfing Between Components

- Surfer jumps between related components and follows signal arrows across drawings.
- Pick a parent and surf to each child, or follow a source arrow to its destinations.
- Surfing is the fastest way to audit that parents, children and signals all agree.



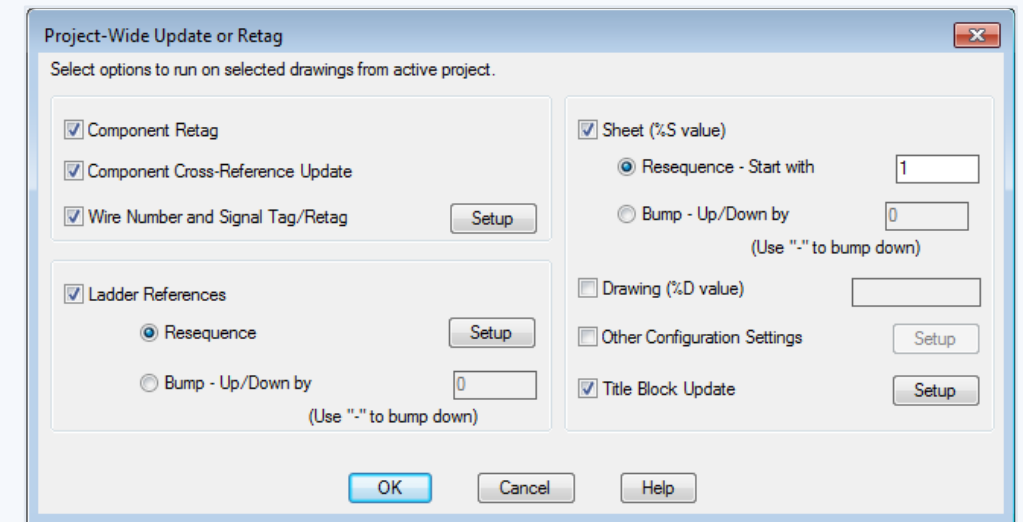
Modify Commands — Edit Without Breaking Intelligence

- Always modify with the Electrical editing commands so component intelligence survives.
- Align components; Move — tags update, wires heal and break as needed.
- Copy with multiple insertion points — tags stay unique; optionally update terminal numbers.
- Circuit Clipboard (Cut/Copy/Paste) moves components and wires between drawings.
- Quick edits: delete-and-heal, toggle NO/NC contact state, swap a block.



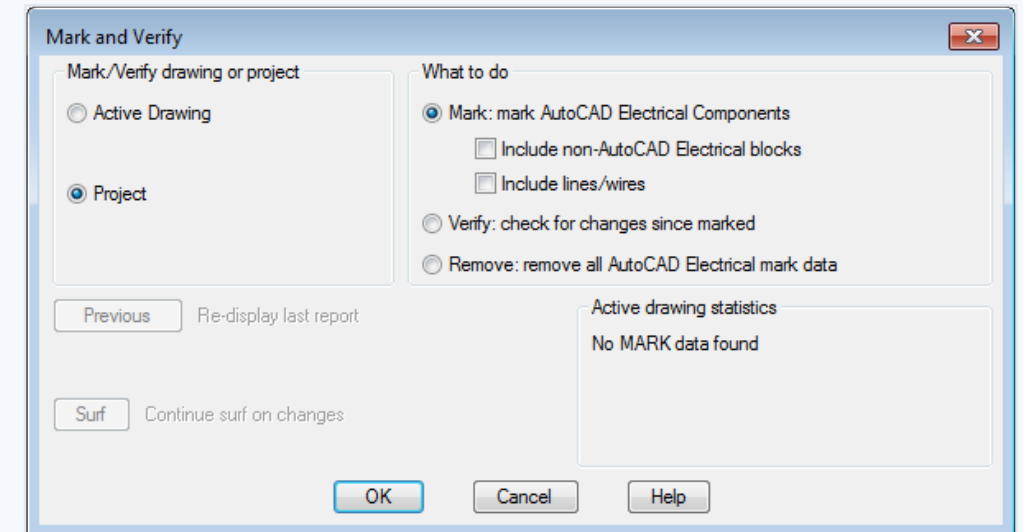
Project-Wide Updates

- Reusing drawings from a previous design? Update the reference system, retag components, reassign wire numbers and refresh cross-references in ONE pass.
- Project-Wide Update/Retag runs across every non-reference drawing in the project.
- This is what keeps a copied design consistent instead of quietly wrong.



Track Changes — Mark and Verify

- Mark/Verify tracks changes to your drawings — even changes made outside AutoCAD Electrical.
- Mark records the current state before edits begin.
- Verify afterwards reports every change — print it or surf straight to each changed object.
- A marked-and-verified set gives the reviewer an exact change list to check against the specification.



Hands-On Labs — Topic 2

Lab 3

- Build Circuits with Circuit Builder
- A schematic sheet containing a Circuit Builder motor circuit plus a saved, reusable circuit block.

Lab 4

- Insert Wires, Wire Numbers and Terminals
- A correctly wired and wire-numbered circuit with terminals, matching the drawing's wire-number specification.

Lab 5

- Cross-Reference and Surf a Drawing Set
- A cross-referenced two-sheet circuit you can navigate with Surfer, plus a Mark/Verify change report.

Build Circuits with Circuit Builder

LAB 3

Use Circuit Builder to insert a three-phase motor-control circuit built dynamically from your selections, then save a frequently used circuit and re-insert it from the icon menu.

You'll build

A schematic sheet containing a Circuit Builder motor circuit plus a saved, reusable circuit block.

Tools: Circuit Builder · Icon Menu · Schematic components · Saved circuits

Build Circuits with Circuit Builder

1

Open the supplied blank sheet 2-0_Blank.dwg inside your TRAINING project and insert a ladder to carry the circuit.

AELADDER

2

Launch Circuit Builder from the Schematic tab and pick a 3-phase motor circuit from the tree.

AECIRCBUILDER

3

Step through the Circuit Configuration dialog — select the disconnect, protection and control options; the circuit is built dynamically from your selections.

Circuit Builder → Configure → Insert

4

Insert a push button and a selector switch from the Icon Menu to complete the control rung.

AECOMPONENT (Icon Menu)

Build Circuits with Circuit Builder

5

Select the finished circuit and save it with Save Circuit to Icon Menu for reuse.

Schematic → Circuit Clipboard → Save Circuit

6

Insert the saved circuit onto a clear area of the sheet and watch tags update to stay unique.

Icon Menu → Saved Circuits

Build Circuits with Circuit Builder

Test it

The motor circuit sits on the ladder with unique component tags (M1, PB1 ...), and your saved circuit re-inserts with automatically retagged components — construction needs are met without redrawing.

Insert Wires, Wire Numbers and Terminals

LAB 4

Insert wires between components, trim them, add automatic wire numbers in the drawing's specified format, and place schematic terminals — the accuracy checks every wiring diagram must pass.

You'll build

A correctly wired and wire-numbered circuit with terminals, matching the drawing's wire-number specification.

Tools: Wires · Wire Numbers · Trim Wire · Schematic Terminals · Power Plan DWG

Insert Wires, Wire Numbers and Terminals

1

Continue on your Lab 3 sheet (keep 10-2_Power_Plan.dwg open as an analysis reference for wiring practice).

OPEN 10-2_Power_Plan.dwg

2

Insert two vertical wires between two horizontal rungs — wires land on the wire layer automatically and tee-dots appear at junctions.

AEWIRE

3

Trim the surplus wire segment with Trim Wire (the connected wires heal).

AETRIM

4

Check the wire-number format on the Drawing Properties → Wire Numbers tab (this is the specification you are wiring to).

AEPROPERTIES → Wire Numbers

Insert Wires, Wire Numbers and Terminals

5

Insert wire numbers project-wide and note that fixed wire numbers are skipped.

`AEWIRENO`

6

Insert two schematic terminals from the Icon Menu (TRMS category) where the wires leave the sheet.

`AECOMPONENT → Terminals`

7

Edit one wire number manually and mark it Fixed, then re-run wire numbering to confirm it is preserved.

`AEEDITWIRENO`

Insert Wires, Wire Numbers and Terminals

Test it

Every wire carries a number in the drawing-property format, the fixed number survives a project-wide renumber, and the terminals join the wires cleanly — the sheet passes an accuracy review.

Cross-Reference and Surf a Drawing Set

LAB 5

Link a parent coil to its child contacts, break a wire across sheets with source/destination signal arrows, then audit the set with Surfer and Mark/Verify — how a checker reviews a real drawing set.

You'll build

A cross-referenced two-sheet circuit you can navigate with Surfer, plus a Mark/Verify change report.

Tools: Cross-referencing · Signal Arrows · Surfer · Mark/Verify · Reference DWG

Cross-Reference and Surf a Drawing Set

1

Insert a relay coil (parent) on sheet 1 and a NO contact (child) on sheet 2 of your project; assign the child to the parent in the Insert/Edit Child dialog.

`AECOMPONENT → coil / contact`

2

Update cross-references and read the parent's contact grid — it reports where every child lives.

`AEXREF`

3

Break a wire network across the two sheets with a Source arrow (assign a code) and a matching Destination arrow.

`AESOURCE · AEDEST`

4

Right-click the coil and Surf: jump between the parent, children and signal arrows from the Surf list.

`AESURF`

Cross-Reference and Surf a Drawing Set

5

Mark the project (Mark/Verify), move one component, then run Verify to get the change report.

```
AEMARKVERIFY
```

6

Open the supplied reference drawings and identify which sheets are reference-only in Project Manager.

```
OPEN 10-  
1_Working_with_References.dwg
```

Cross-Reference and Surf a Drawing Set

Test it

Surfer jumps coil → contact → signal arrow across sheets without manual searching, and the Verify report lists exactly the one component you moved — the set is auditable for accuracy.

Recap — Topic 2

- A4 — analyse drawings to determine construction needs; assemble standard motor-control circuits (Lab 3)
- A3 — review drawings for accuracy; wire the circuit and number it to specification (Lab 4)
- A3, A4 — review a multi-sheet set for accuracy with cross-references, Surfer and signal arrows (Lab 5)
- Key concepts: Types of technical drawings, Technical drawing standards, Schematic components, Terminals & PLC symbols, Wires, numbers & arrows, Review for accuracy.

SEE YOU TOMORROW · 0930

End of Day 1

Day 2 continues with drawing-set review, documentation, submission — and the final assessment.

Day 2

Digital attendance (AM) · Day 1 recap

TOPIC 03

Reviewing Technical Drawings for Strategic Relevance and Standard Compliance

Drafting documentation · panel layouts · reports & part lists · custom symbols · plot & publish for submission (K5, K6 · A5, A6)

Key Concepts — Topic 3

1

Fundamentals of documentation

Panel layouts, nameplates, item numbers and balloons turn a schematic into build documentation. (K5)

2

Reports & part lists

BOM, wire from/to and component reports extract project data for checking and procurement.

3

Custom symbols

Symbol Builder creates house-standard symbols so drawings match the organisation's conventions. (K6)

4

Strategic relevance

Reuse projects, track changes and align drawing sets with the organisation's design strategies. (A5)

5

Standard compliance

Title blocks, xrefs and plot standards keep sheets conformant to submission requirements. (A6)

6

Plot & publish

Output the project to a printer or a hyperlinked multi-sheet PDF for stakeholder submission.

Fundamentals of Drafting Design Documentation

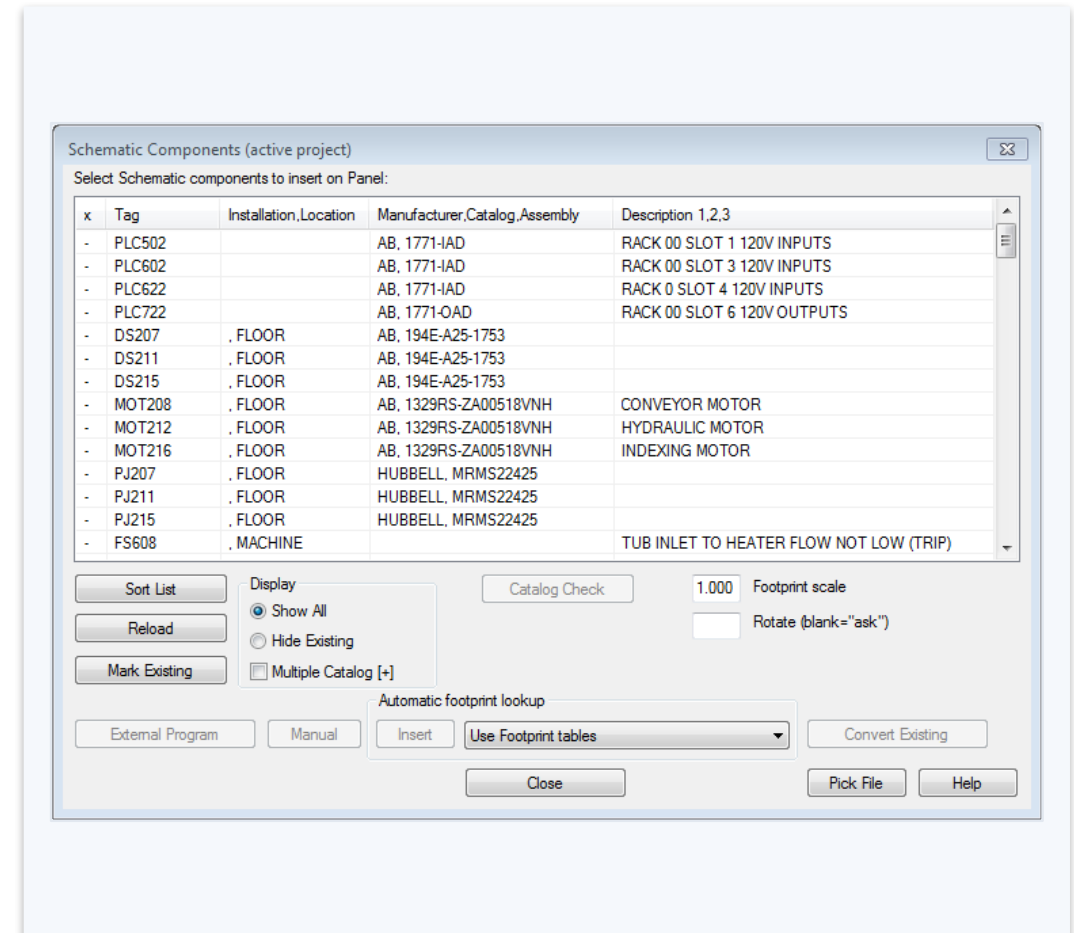
- A schematic alone cannot be built: documentation adds the physical and procurement views.
- Panel layouts show where each device mounts; nameplates label them; item numbers and balloons key parts to the BOM.
- Reports extract part lists, wire lists and terminal plans from the same project data — no retyping.
- Documentation must align with the organisation's strategies (house symbols, formats) and submission standards. (A5, A6)

Panel Layouts

- Insert panel footprint symbols from the schematic component list or the panel icon menu.
- The PANEL ribbon tab carries the layout tools; drawing properties hold panel-specific settings.
- Add components and connect with the insert tools; nameplates, item numbers and balloons complete the layout.

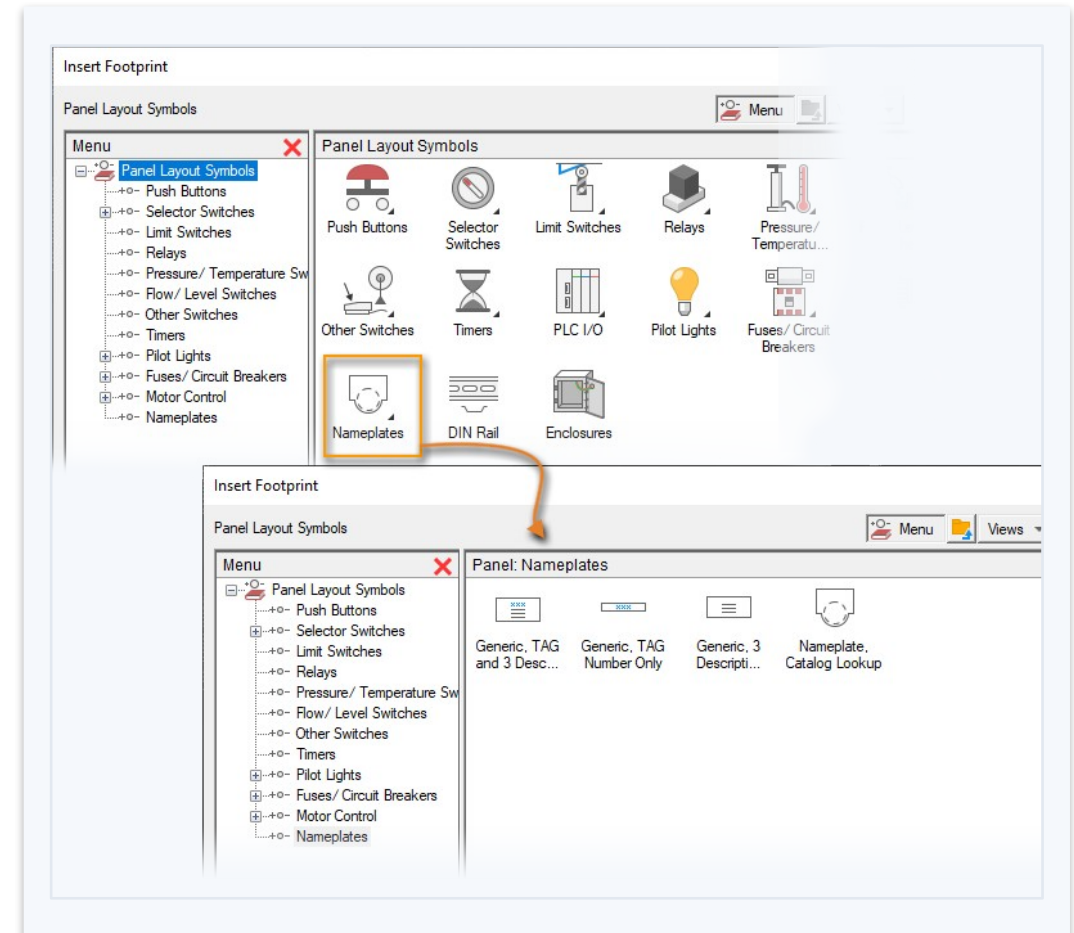
Insert Footprints from the Schematic List

- Draw schematics first, then build the panel by selecting from the schematic component list.
- The catalog assignment finds the correct footprint; browse or draw one on the fly if missing.
- The dialog shows which components are already placed; all values copy from schematic to footprint.
- Change a value on either side later and the other side updates — schematic and panel stay consistent.



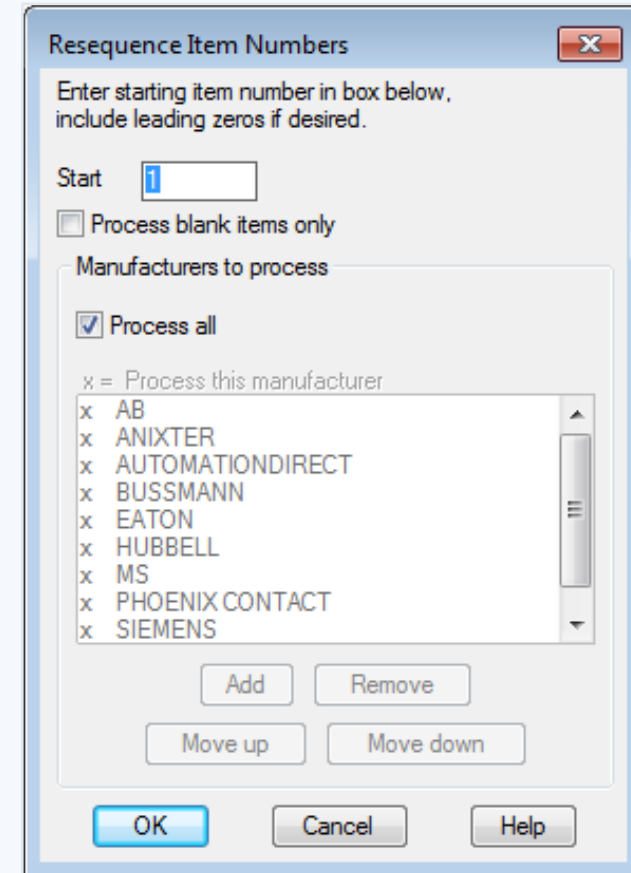
Panel Icon Menu and Nameplates

- Components not on the schematic (wire duct, rails) insert from the Panel Icon Menu.
- Insert nameplates from the icon menu and associate each with a footprint.
- Tag, installation, location and descriptions copy from the footprint — override if required.



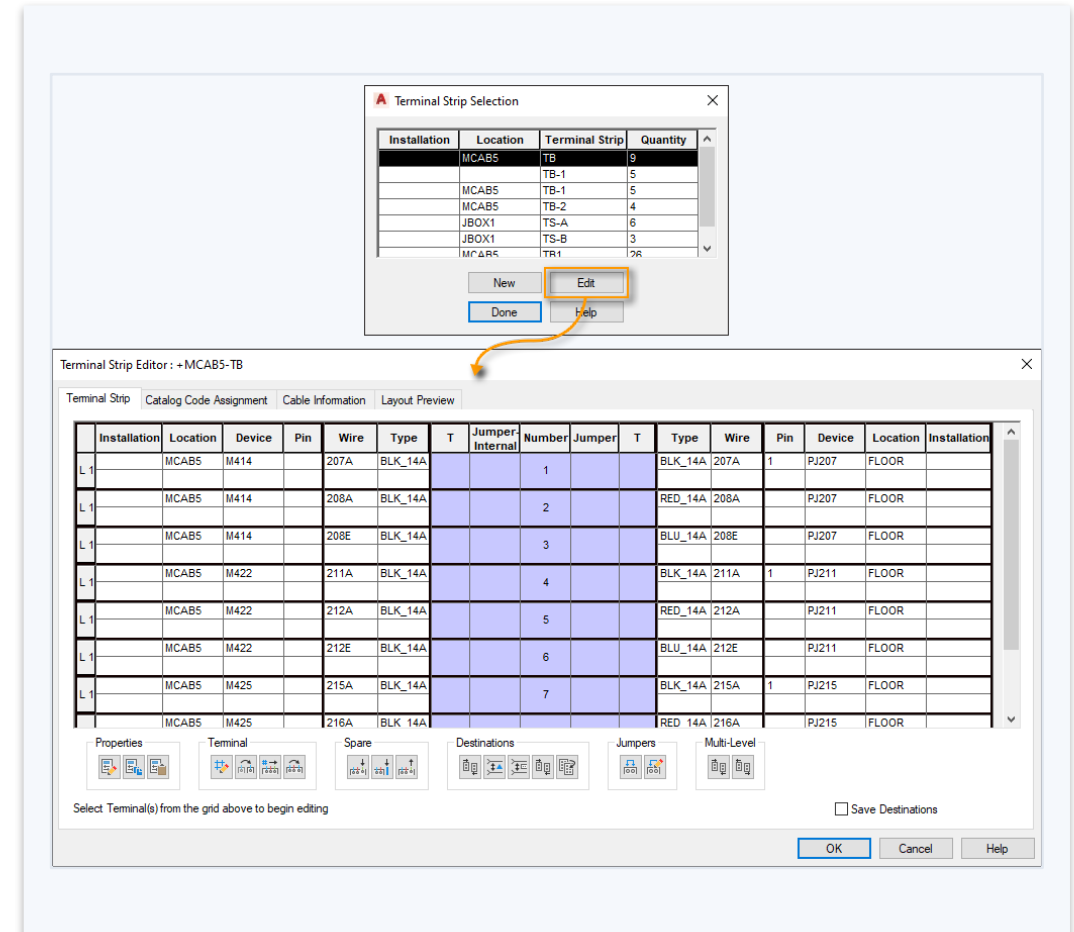
Item Numbers and Balloons

- Item numbers key parts to the BOM; components sharing a catalog value share an item number.
- Panel components are numbered first, then schematic components.
- Set the item-numbering project properties before resequencing.
- Insert balloons to make item numbers visible; multi-catalog components prompt for which value to balloon.



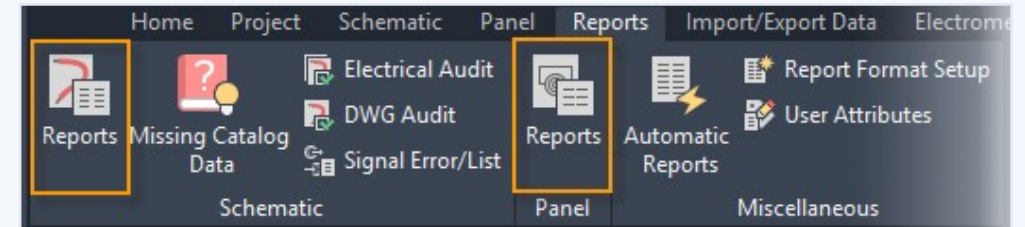
Terminal Strip Editor

- Manage an entire terminal strip at once: catalog values, connected components, wires and numbers.
- Resequence terminal numbers, reorder terminals, add spares and accessories, insert jumpers.
- The strip list merges panel terminals and schematic terminals into one editable view.
- Insert the strip graphically — or generate tabular terminal strips with the Table Generator.



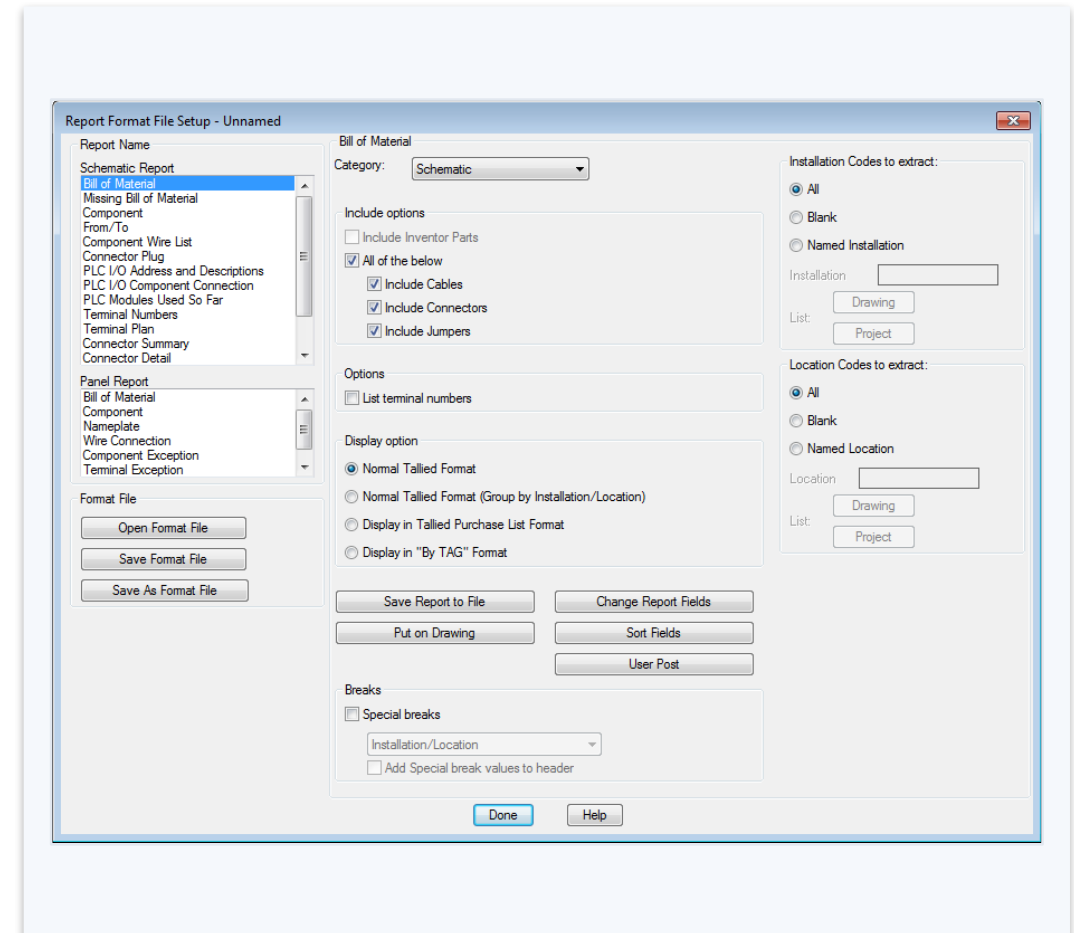
Reports

- Generate project reports: Bill of Material, component lists, wire from/to, terminal plans and more.
- The REPORT command builds custom reports from drawing data; select the report, options and drawings to process.
- Output to a file for procurement, or insert on the drawing as a table for the construction set.



Report Fields and Format Files

- After running a report, include/exclude fields, reorder them and rename labels.
- Save the settings as a .SET format file — unlimited format files per report type.
- Format files ARE the organisation's reporting standard: everyone reproduces the same layout in one click.
- Automatic Reports runs a whole group of formatted reports in a single pass.



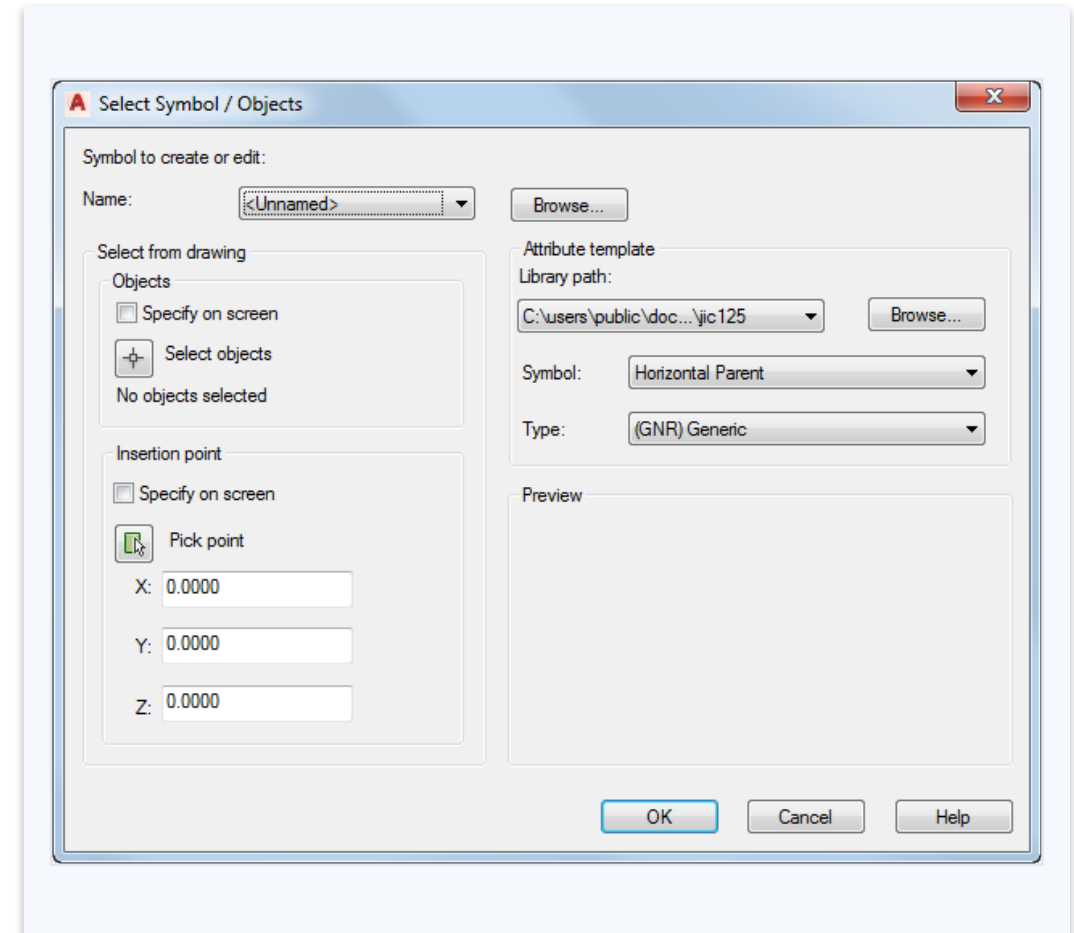
Output a Report — File or Drawing Table

- Save to File exports the report (CSV, XLS, MDB) for procurement and checking.
- Put on Drawing inserts the report as an AutoCAD table — style it with TABLESTYLE.
- The drawing table and the exported file come from the same data, so they cannot disagree.

Report for Project VDDEMO					
TAGS	QTY	SUB	CATALOG	MFG	DESCRIPTION
CR406 CR407 CR408	3		700-P200A1	AB	P TYPE ELECTRICALLY HELD RELAY, 4 POLE, AC COIL, 115-120VAC 60Hz / 110VAC 50Hz COIL, RATING 10A, OPEN TYPE P 120VAC 2 NO
PB414	1	*1	800EP-E2	AB	PUSH BUTTON - MOMENTARY, IP66, NEMA 4/4X/13 22.5mm EXTENDED, IEC STYLE BLACK 2 NO PLASTIC OPERATOR w 2 ACROSS MTG
		*2	800E-2X10	AB	800E CONTACT BLOCK 22.5mm IEC STYLE 1 NO
		*1	800E-A2L	AB	800E MOUNTING LATCH - 2 ACROSS 22.5mm IEC STYLE 2 ACROSS MOUNTING LATCH 2 ACROSS MOUNTING LATCH
LT415 LT423 LT426	3		800H-PR16G	AB	GREEN PILOT LIGHT - STANDARD, ROUND DILTIGHT 30.5mm 120VAC XFMR PLASTIC LENS CORROSION RESISTANT
LT409 LT411 LT412 LT413	4		800H-PR16R	AB	RED PILOT LIGHT - STANDARD, ROUND DILTIGHT 30.5mm 120VAC XFMR PLASTIC LENS CORROSION RESISTANT
SS406 SS433	2		800MR-HH2BB	AB	SELECTOR SW - 2 POS MAINT, NEMA 13 22.5mm BLACK KNOB 2 NO 2 NC

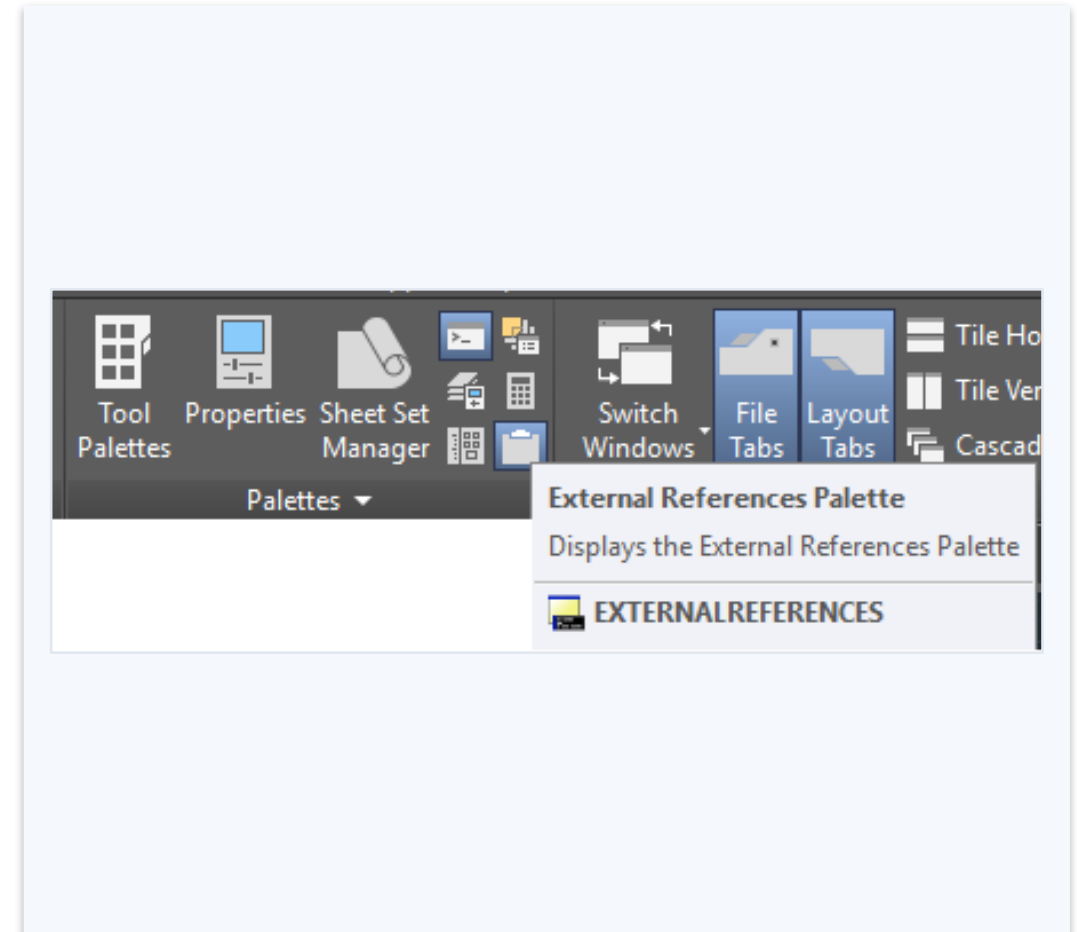
Custom Symbols with Symbol Builder

- Symbol Builder creates custom symbols fully compatible with AutoCAD Electrical.
- Start from scratch or from a similar existing symbol; set symbol type, insertion point and orientation.
- For schematic symbols create both the parent and child versions.
- Draw the graphics, then save following the AutoCAD Electrical symbol naming convention — that is what makes the symbol intelligent.



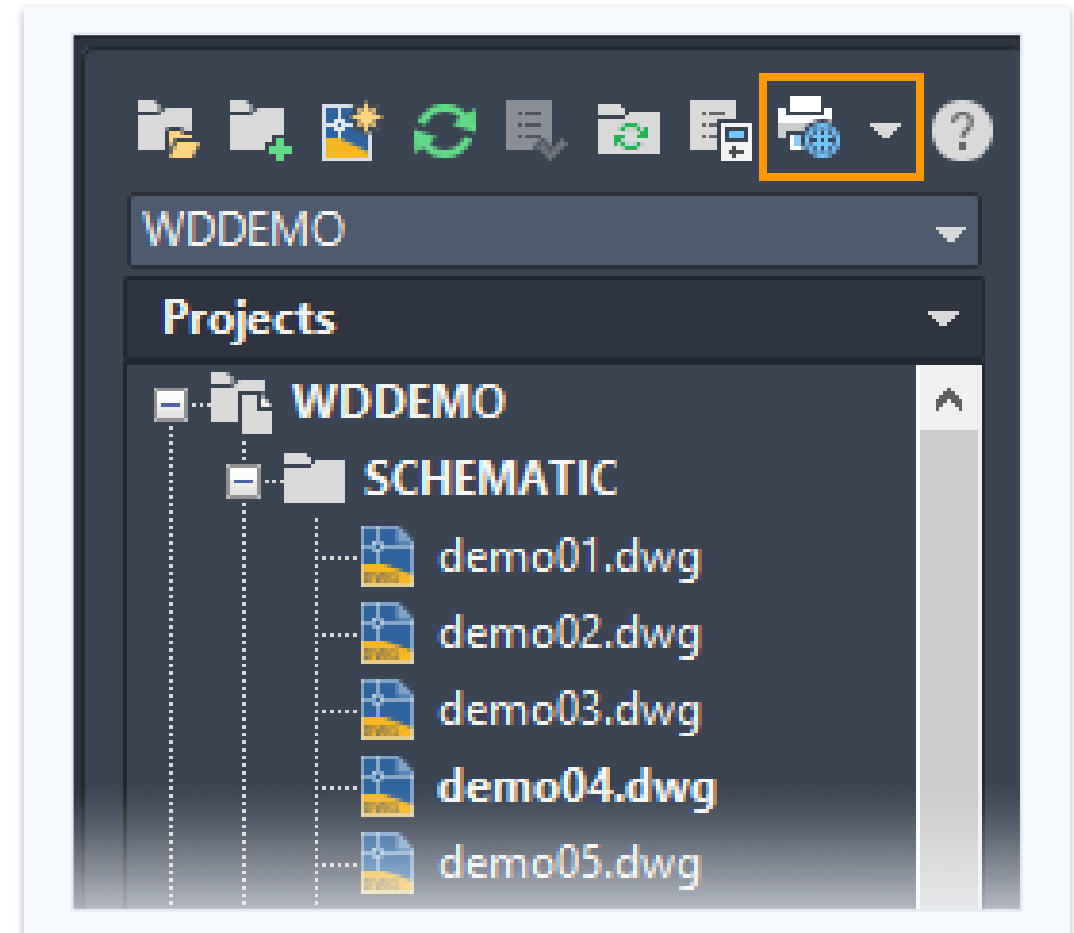
External References (Xrefs)

- An external reference attaches another file into the current drawing without copying it.
- The External References palette organises DWG, DWF, DGN, PDF and image references.
- Attach via the palette (View tab → Palettes) or the XREF command; pick the file and placement.
- Xrefs keep shared content (title blocks, outlines) in ONE file — edit once, update everywhere.



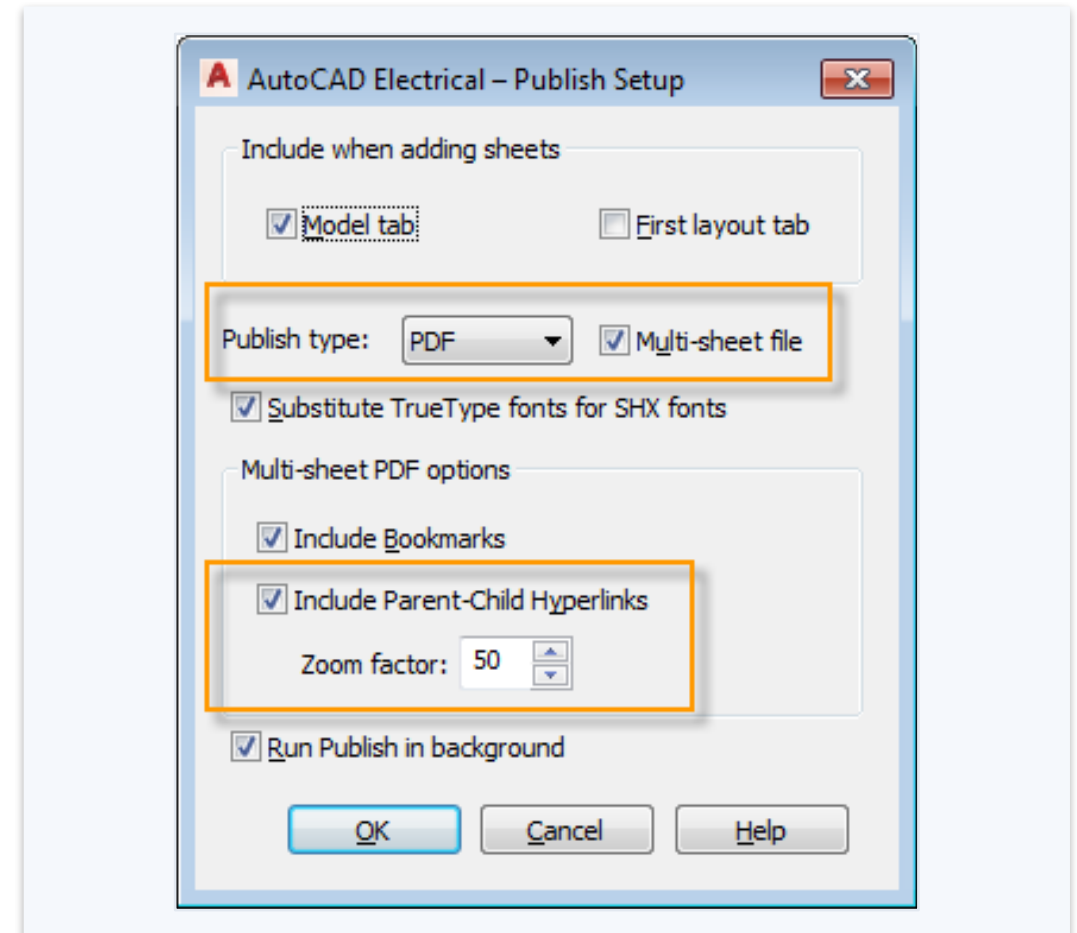
Plot a Project

- Print an entire project, a subset of sheets, or a single drawing.
- Use Publish/Plot on Project Manager → Plot Project, then pick the drawings to output.
- Select the layout to plot per drawing; use a named page setup for consistent output.
- Map a plot date/time attribute in the title block — it stamps automatically at plot time.



Publish to a Hyperlinked PDF

- Publish the project to a multi-sheet PDF that preserves relationships as hyperlinks.
- Tick Multi-sheet file and Include Hyperlinks; the zoom factor sets how close a clicked link zooms.
- Each object carries one hyperlink — reach a child through the parent's cross-reference text.
- Update cross-referencing BEFORE publishing; then reviewers can navigate the set without AutoCAD.



Hands-On Labs — Topic 3

Lab 6

- Create a Panel Layout with Footprints
- A panel layout drawing with footprints placed from the schematic list, a nameplate, item numbers and balloons.

Lab 7

- Generate Reports and Part Lists
- A BOM table placed on the drawing plus a saved .SET format file and a CSV export of the report.

Lab 8

- Plot and Publish the Project to PDF
- An A3 PDF plot of one sheet and a hyperlinked multi-sheet PDF of the whole project.

Create a Panel Layout with Footprints

LAB 6

Generate a panel layout from the schematic component list, insert footprints and a nameplate, and assign item numbers with balloons — drafting design documentation the workshop can build from.

You'll build

A panel layout drawing with footprints placed from the schematic list, a nameplate, item numbers and balloons.

Tools: Panel tab · Schematic List · Footprints · Nameplates · Item balloons · Blocks DWGs

Create a Panel Layout with Footprints

1

Create a new panel drawing in your project and switch to the Panel tab of the ribbon.

New Drawing → Panel tab

2

Insert footprints from the schematic list — AutoCAD Electrical matches each component's catalog value to a footprint.

AEFOOTPRINT (Schematic List)

3

Place the footprints inside the panel outline; values (tag, location, description) copy over from the schematic.

Pick insertion points

4

Insert a nameplate from the Panel icon menu and associate it with a footprint.

Panel Icon Menu → Nameplate

Create a Panel Layout with Footprints

5

Assign item numbers project-wide — components with the same catalog value receive the same item number.

`AEPANELITEM`

6

Add item balloons to the footprints and study the supplied blocks/attributes drawings to see how footprint blocks carry data.

`AEBALLOON`

Create a Panel Layout with Footprints

Test it

Each footprint shows the same tag as its schematic parent, the nameplate carries the footprint's description, and balloons display consistent item numbers — the panel documents the design faithfully.

Generate Reports and Part Lists

LAB 7

Run Bill of Material and Wire From/To reports across the project, tailor the fields, save a format file for repeat use, and insert the report as a table on the drawing — the part lists a checker reviews.

You'll build

A BOM table placed on the drawing plus a saved .SET format file and a CSV export of the report.

Tools: Reports · Bill of Material · Format files (.SET) · Tables · Attributes DWG

Generate Reports and Part Lists

1

Run the Reports tool on the Reports tab and choose Bill of Material, processing the whole project.

AEAUDIT → Reports → BOM

2

Adjust the report: include/exclude fields, reorder columns and rename field labels to your house standard.

Change Report Fields

3

Save the settings as a format file so the same report can be re-run in one click next time.

Save Format File (.SET)

4

Insert the report on the drawing as a table (Put on Drawing) using a table style.

Put on Drawing · TABLESTYLE

Generate Reports and Part Lists

5

Export the same report to a file for procurement (CSV/XLS).

Save to File → CSV

6

Run a Wire From/To report and compare it against the wiring you created in Lab 4; open the supplied table drawings to see linked tables.

Reports → Wire From/To

Generate Reports and Part Lists

Test it

The BOM table on the drawing matches the CSV export line for line, and re-running the report through your .SET format file reproduces the same layout — the part list is standard-compliant and repeatable.

Plot and Publish the Project to PDF

LAB 8

Attach an external reference, then plot a drawing to an A3 PDF and publish the whole project to a single multi-sheet PDF with hyperlinks — the final conformance step before submission.

You'll build

An A3 PDF plot of one sheet and a hyperlinked multi-sheet PDF of the whole project.

Tools: External References · Plot · Publish · Multi-sheet PDF · Layout tabs DWG

Plot and Publish the Project to PDF

1

Open the External References palette and attach a DWG as an xref to see how reference files are managed.

XREF → Attach DWG

2

Open 1-12_Using_Layout_Tabs.dwg and review its layout tabs — plotting always happens from a layout with a page setup.

OPEN → Layout tabs

3

Plot the active drawing to PDF at A3 size from the Plot dialog (printer: AutoCAD PDF, paper: ISO A3).

PLOT → AutoCAD PDF → ISO A3

4

From Project Manager, choose Publish/Plot → Plot Project and select the sheets to output.

Project Manager → Plot Project

Plot and Publish the Project to PDF

5

Publish the project to a single multi-sheet PDF with Include Hyperlinks checked — cross-references become clickable links.

AEPUBLISH → Multi-sheet · Hyperlinks

6

Open the PDF and click a parent's cross-reference to jump to its child; confirm the title block shows the plot date.

Verify in PDF viewer

Plot and Publish the Project to PDF

Test it

The A3 PDF matches the layout exactly, and in the multi-sheet PDF clicking a cross-reference hyperlink jumps to the referenced component — the set is ready for standards-conformant submission.

Recap — Topic 3

- A5 — turn the schematic into build documentation aligned with the organisation's design strategy (Lab 6)
- A6, K5 — extract BOM and wire lists, format them to house standards and place them on the drawing (Lab 7)
- A6 — output the drawing set to submission standards (A3 sheets, hyperlinked multi-sheet PDF) (Lab 8)
- Key concepts: Fundamentals of documentation, Reports & part lists, Custom symbols, Strategic relevance, Standard compliance, Plot & publish.



WRAP-UP

Course Summary & Assessment

What You Achieved

1

Set up to standard

Configured drawings from templates with limits, units, grid and snap — production metrics in force. (A1)

2

Projects & drawings

Built .wdp projects, managed drawing order, properties and reference-only sheets. (A1, A2)

3

Schematic drawing

Inserted components, circuits, terminals and PLC modules to IEC/IEEE/NFPA standards. (K1, K3)

4

Wired & numbered

Wired circuits with layered wires, spec-format wire numbers and signal arrows. (K4, A3)

5

Reviewed drawing sets

Audited accuracy with cross-references, Surfer and Mark/Verify. (A3, A4)

6

Documented & submitted

Produced panel layouts, BOM reports, custom symbols and hyperlinked PDF submissions. (K5, K6, A5, A6)

Practice Exam

- Sharpen your readiness with the Tertiary Infotech practice exam portal.
- Practice exams: <https://exams.tertiaryinfotech.com/>
- Attempt it under timed conditions and review every explanation.
- Revisit any lab whose topic you miss, then re-attempt.

Recommended Courses

1

WSQ - Technical Drawing with AutoCAD

2

WSQ - Technical Drawing with AutoCAD
Mechanical

3

WSQ - Application of BIM using Revit

4

WSQ - Architecture Drawing with Revit

5

WSQ - Product Design with Fusion 360

6

WSQ - AutoCAD Civil 3D for Infrastructure
Design

Support

- If you have any enquiries during and after the class, contact us:
- Email: enquiry@tertiaryinfotech.com
- Tel: +65 6100 0613
- Website: www.tertiarycourses.com.sg

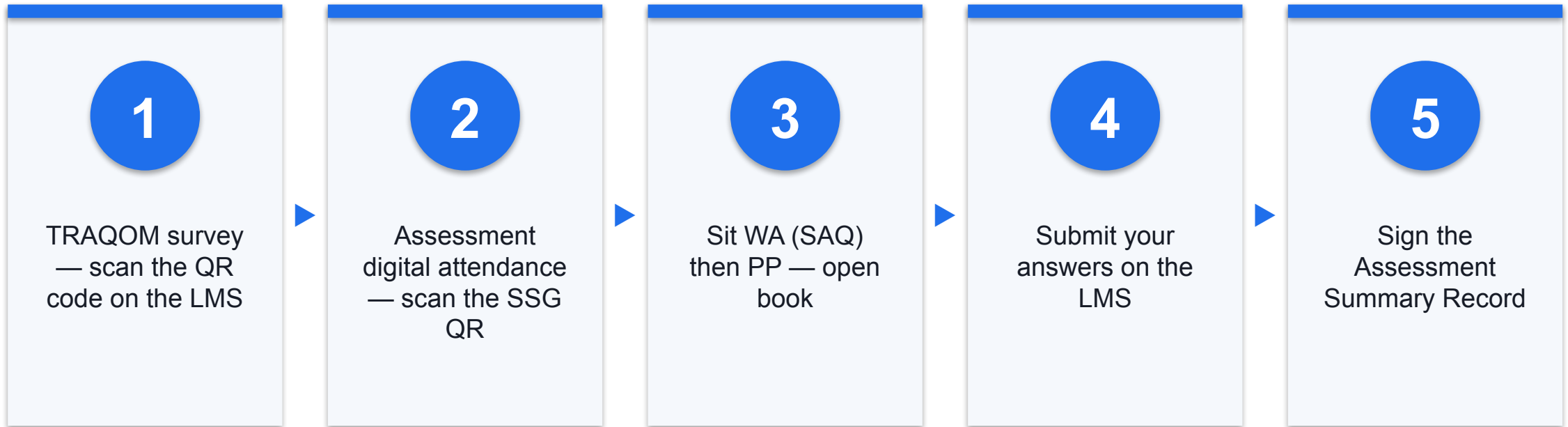
Cert & TRAQOM Survey (Mandatory)

- Complete the course feedback and the mandatory SSG TRAQOM survey.
- Your e-certificate is issued via the LMS/TMS portal: <https://lms-tms.tertiaryinfotech.com>
- The trainer will display the TRAQOM QR code — please complete it before the assessment.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book. (K1–K6)
- Practical Performance (PP) — hands-on AutoCAD Electrical tasks, 1 hour, open book. (A1–A6)
- Format: Open Book — slides, Learner Guide and approved materials only.
- Submit your completed scripts to the assessor and sign the Assessment Summary Record.

Assessment Flow



Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

HAPPY DRAFTING

Thank You!

You are now ready to produce, analyse and review technical drawings with AutoCAD Electrical.